

Bayesian Structural Time Series Analysis of Temperature Extremes

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Abstract

Return periods are no longer constants in a changing climate: exceedance risk becomes a trajectory that should be monitored, updated, and forecast with uncertainty. Leveraging asymptotic theory as far as long environmental records allow, we develop a structural Dynamic GEV (DGEV) model by embedding block extremes in a probabilistic state-space framework that mirrors the basic structural models used for Gaussian time series. This yields a unified yet separable parametric view of bulk and tails: the same level-slope-seasonality decomposition can be tracked for both averages and extremes, producing posterior trajectories of time-varying hazard, return periods, and high quantiles for decision-relevant questions. Bayesian computation is challenging in the slowly varying climate background regime, where evolution variances can be near zero whereas natural variability is orders of magnitude bigger. We address this with a reparametrisation of the state space and global-local shrinkage priors, and make the non-Gaussian DGEV practical via an approximate latent-state update based on a fast local Laplace step. We then apply the framework to a long record of daily temperatures from Uccle summarized into six monthly indices, to compare mean and extreme changes. We demonstrate posterior risk tracking for societally relevant thresholds, derive calibrated posterior predictive distributions for future extremes, propose goodness-of-fit diagnostics, and show how the model explains the sudden emergence of record-breaking heat events under modest warming.

Keywords: Extreme Value Theory, MCMC, State Space Models, Climate Change, Bayesian Structural Time Series

1 Introduction

“Stationarity is dead” (Milly et al., 2008): as the climate system evolves, the assumption that extremes are independent replicates from a fixed distribution is no longer tenable for risk assessment. What is needed instead are time-series methods for extreme events that treat hazard as time-varying and serially dependent, making exceedance risk and related design metrics trackable (and forecast-able) through time (Slater et al., 2020). Such analyses must be coherently probabilistic to support decision-making under uncertainty (Salas and Obeysekera, 2014). Even in slowly varying climates, modest distributional shifts can materially alter far-tail risk and the expected impact of rare events, so inference should deliver time-varying exceedance probabilities and (revised definitions of) return levels and periods (Rootzén and Katz, 2013; Cooley, 2012) in order to prepare for a reconfigured hazard landscape.

Recent disaster compilations underscore the scale of societal exposure to these low-probability hazards: (UNDRR, 2020) documents a marked rise in major recorded natural hazard-related disaster events in 2000–2019 relative to the preceding two decades, while EM-DAT reports 7348 recognized disasters over the same period, claiming 1.23 million lives and affecting 4.2 billion people, with approximately € 2.72 trillion in economic losses (UCLouvain, 2024). Combining event attribution with socio-economic damage data, Newman and Noy (2023) estimate that at least 143 billion USD per year of extreme-event losses in the United States are already directly attributable to anthropogenic climate change, consistent with evidence that the recent surge is primarily associated with more frequent extreme meteorological events such as floods, storms, heatwaves and droughts (Donat et al., 2013). Looking ahead, projections indicate more frequent and severe heatwaves, heavy rainfall and droughts in many regions (Easterling et al., 2000; Fischer et al., 2013), and detection and attribution studies indeed increasingly quantify a substantial human contribution to these extremes (Easterling et al., 2016; Seneviratne et al., 2021; Hegerl et al., 2004; Wang et al., 2017; Zwiers et al., 2011; Robinson et al., 2021; Engdaw et al., 2023). Although Milly et al. (2008) argued from water-resources design, the point extends broadly to environmental extremes: climate-related risk must urgently be accounted for in design and planning under changing hazard (Data, 2009; Keller et al., 2021). In this paper we focus on temperature.

Global temperature extremes have changed markedly over the last century, with more frequent warm extremes and fewer cold extremes (Alexander et al., 2006; Alexander, 2016), against a background of roughly 1.1°C global mean surface warming since pre-industrial times (IPCC, 2023). Observation-based syntheses based on ETCCDI (Expert Team on Climate Change Detection and Indices (ETCCDI), 2024) daily extreme indices show widespread, spatially coherent signals consistent with warming, typically strongest for minimum-temperature indices: many land regions exhibit fewer cool nights and more warm nights, alongside a reduced diurnal temperature range (Donat and Alexander, 2012; Donat et al., 2013; Perkins et al., 2012). Climate-model evidence supports the same broad separation between shifts in the bulk and shifts in the tails: the evolution of means and extremes can differ regionally (Kharin and Zwiers, 2005), and cold extremes tend to warm faster than warm extremes, especially in mid-high latitudes where snow and sea-ice retreat amplifies winter warming

(Kharin et al., 2007, 2013; Li et al., 2021). A fundamental challenge for nonstationary risk assessment is therefore that tail risk can change disproportionately to the bulk: even modest shifts in the mean or variability can induce large changes in far-tail probabilities (Schär et al., 2004; Katz and Brown, 1992; Huybers et al., 2014; McKinnon et al., 2016). Quantile-based analyses confirm that distributional change in temperature is not uniform: tail behavior can depart strongly from shifts in the bulk (Haugen et al., 2018) under time evolution and can be asymmetric across quantiles and seasons (Matiu et al., 2015; Dunn et al., 2019). However, while such non-parametric tools are well suited to well-sampled quantiles, they are not designed for inference at very small exceedance probabilities in the far tail (Chernozhukov et al., 2017)—a requirement for assessing unprecedented events (e.g. $\geq 40^\circ\text{C}$ days in western Europe) now and under future conditions (Vautard et al., 2020). In contrast, extreme value theory (EVT) targets rare-event behaviour directly and provides a principled framework for tail inference and extrapolation beyond the observed range (Coles et al., 2001).

Classical EVT applications in climate and hydrology often assume stationarity and fit Generalized Extreme Value (GEV) or Generalized Pareto Distribution (GPD) models to block maxima or peaks over threshold, sometimes only after a priori detrending (Mentaschi et al., 2016); yet this prewhitening practice blurs potential different non-stationary behaviour in the mean and in the tail as we emphasized, so the field has been shifting toward handling nonstationarity within the EVT framework itself (Katz et al., 2002; Katz, 2013). Accordingly, most nonstationary EVT models let GEV/GPD parameters depend on time-varying covariates (Wang et al., 2017; Zwiers et al., 2011; Kim et al., 2016; Wang et al., 2004), on time (typically linear) (Kharin and Zwiers, 2005; Kharin et al., 2013; Cheng et al., 2014), or occasionally higher-order polynomials (Katz, 2013; Caroni and Panagoulia, 2016), including combinations (Robin and Ribes, 2020; Brown and Ferro, 2008) and more ad-hoc functions (Contzen et al., 2023; Um et al., 2017). Much of this literature still focuses on annual maxima/minima (Kharin and Zwiers, 2000, 2005; Kharin et al., 2007) (or on anomalies after “bulk”-deseasonalizing; e.g. Brown and Ferro (2008)), and then injects flexibility either through moving-window fits (Kharin et al., 2013; Zhang et al., 2017; Bhatia and Ganguly, 2019) or via changepoint models (Beaulieu et al. (2012, 2024); Cahill et al. (2015); Rodionov (2004) and more specifically in EVT, Le Roux et al. (2022)). Both are shaky and restrictive for statistically valid risk quantification: moving windows recycle strongly dependent data and can smear actual regime shifts into artificial “ramps”, while changepoint models hinge on break-number/placement choices (for regime shifts) and multiple-testing; and don’t allow for smoothly varying behaviour (Rahmstorf et al., 2017).

Low-dimensional nonstationary EVT specifications (e.g. linear time trends or linear covariate links) are attractive because they are easy to interpret, communicate and benchmark, but when hazard evolves beyond a fixed deterministic form one needs a genuinely dynamic approach. Such a dynamic, parametric model should (i) retain an extreme-value observation model (to move beyond the well-sampled quantiles), (ii) represent serial dependence explicitly, and (iii) accommodate slowly evolving trends and seasonal structure within a flexible, modular framework (without

prescribing them in a deterministic sense or removing it by pre-whitening) that supports coherent uncertainty quantification of risk in past, present and future. To this end we adopt a structural state-space viewpoint in which the distributional location is decomposed into a few interpretable components—a local level, slope, and seasonal amplitudes—whose temporal smoothness is governed by the innovation variances of a Markov chain (West and Harrison, 2006; Durbin and Koopman, 2012; Petris et al., 2009). While structural state-space models were originally developed for Gaussian data (Harvey, 1990) (and more generally for exponential-family observations; West et al. (1985); Helske (2017)) the same structural decomposition naturally carries over to extremes by simply replacing the observation layer with a GEV model for block maxima, yielding the Dynamic GEV (DGEV) formulation (Gaetan and Grigoletto, 2004). Gaetan and Grigoletto (2004) already recognized that fully parametric time trends in GEV parameters are overly restrictive and proposed this semi-parametric state space alternative (though not in the structural time series parametrization) for smoothing dynamic extremes. In their formulation however, innovation variances are fixed ‘roughness penalties’ rather than learned parameters, so the latent process is in fact used mainly as a smoothing device for block maxima rather than as a probabilistic time-series model that supports coherent uncertainty quantification and genuine forecasting. Huerta and Sansó (2007) take an import step for the DGEV toward fully Bayesian learning by placing inverse-gamma priors on the evolution variances, but do so through a specification that effectively introduces model misspecification to retain a convenient conjugate structure (see Section 2), which casts doubt on its status as a genuine probabilistic state-space model in the sense of Särkkä (2013). Moreover, inverse-gamma priors are notorious in the small-variance regime relevant for slowly varying climate dynamics (Gelman, 2006; Frühwirth-Schnatter and Wagner, 2010). We experienced this ourselves when implementing the MCMC framework of Huerta and Sansó (2007) on a long temperature record. First, posterior learning of the evolution variances proved indeed highly sensitive to the chosen hyperparameters. Second, in slowly varying climatic regimes the resulting near-zero innovation scales induced extreme trajectory-scale coupling and very high autocorrelation in the Markov chain—an issue now well understood as a parametrisation (centering vs non-centering) problem in hierarchical models (Papaspiliopoulos et al., 2007) which makes inference fragile and prohibitively slow. We therefore draw on recent developments from the econometric time-varying-parameter (TVP) literature and adopt shrinkage priors that behave well near the origin (Belmonte et al., 2014; Bitto and Frühwirth-Schnatter, 2019; Cadonna et al., 2020) together with the structural time-series reparametrisation of Frühwirth-Schnatter and Wagner (2010), which restores efficient mixing.

Section 2 introduces the structural state-space model for extremes as well as averages, and Section 2.1 develops its non-centred reparametrisation. Posterior computation is then constructed in Section 3 via a Gibbs scheme alternating between latent-state and parameter updates: fully conjugate for Gaussian observations (under suitable priors), and for DGEV made practical by a fast local Laplace approximation (Appendix C) that yields efficient approximate state updates alongside partially

conjugate parameter steps. Section 4 formalizes return periods under nonstationarity in terms of time-varying exceedance probabilities, and Section 5 exploits the state-space structure to deliver forecasts that propagate observational and dynamic uncertainty into the future. Goodness-of-fit and predictive performance are then assessed in Section 6 using probability integral transform (PIT) diagnostics and expanding-window cross-validation. We apply the framework in Section 8 to daily temperature measurements from Belgium (1892–2022), described in Section 7, aggregated to monthly extremes and averages, to obtain a unified parametric description of distributional change and evolving warm/cold risk for day- and night-time temperatures. The resulting analysis moves beyond deterministic trend fitting, discrete changepoint formulations for changes in warming rates, and bulk-based a priori smoothing, in the sense that it infers a full posterior over continuously evolving hazard curves that can be updated sequentially as new observations arrive. In Section 9 we translate these trajectories into decision-relevant examples tied to concrete thresholds (heat records) and standard ETCCDI indices, and we clarify how nonstationarity interacts with bounded tails and the implication for risk.

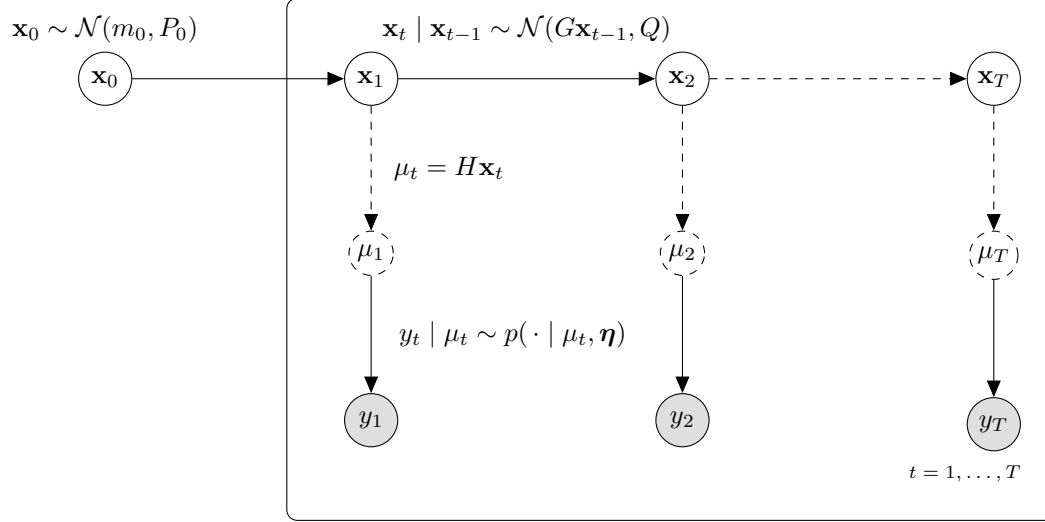


Fig. 1: Structural state-space model with latent Gaussian dynamics and a (possibly non-Gaussian) observation density $p(\cdot)$. The transition layer is governed by (G, Q) , while the observation layer links the state to the location parameter via $\mu_t = H\mathbf{x}_t$ and uses static observation parameters $\boldsymbol{\eta}$. The dashed μ_t nodes are shown for illustration (they are deterministic functions of $\mathbf{x}_t$) but are not part of the state space PGM.

2 Structural time series

For a univariate time series $\{y_t\}_{t=1}^T$ with seasonal period s , we consider the structural state space model (see e.g. (Särkkä, 2013)) with a latent Gaussian Markov structure and an observation density $p(\cdot)$:

$$\mathbf{x}_0 \sim \mathcal{N}(\mathbf{m}_0, P_0), \quad (1)$$

$$\mathbf{x}_t | \mathbf{x}_{t-1} \sim \mathcal{N}(G\mathbf{x}_{t-1}, Q), \quad t = 1, \dots, T, \quad (2)$$

$$y_t | \mathbf{x}_t \sim p(y_t | \mu_t, \boldsymbol{\eta}), \quad \mu_t = H\mathbf{x}_t. \quad (3)$$

Here $(\mathbf{m}_0, P_0)$ parameterize the Gaussian initial state distribution, Q is the innovation covariance matrix of the latent evolution, and $\boldsymbol{\eta}$ collects the (static) parameters of the observation density. The design row H maps the latent state $\mathbf{x}_t$ to a distributional location parameter μ_t . The transition matrix G and covariance Q together encode the structural decomposition and the degree of temporal smoothness. The resulting probabilistic graphical model is shown in Figure 1. To make this explicit, we decompose the location into a trend and a seasonal component,

$$\mu_t = \alpha_t + \gamma_t, \quad (4)$$

and define the state vector as

$$\mathbf{x}_t = \begin{pmatrix} \alpha_t \\ \beta_t \\ \gamma_t \end{pmatrix} \in \mathbb{R}^{s+1}, \quad \gamma_t = (\gamma_t, \gamma_{t-1}, \dots, \gamma_{t-s+2})^\top. \quad (5)$$

With this convention,

$$H = (1, 0, 1, 0, \dots, 0),$$

so that $H\mathbf{x}_t = \alpha_t + \gamma_t$. The state transition matrix can be written as the direct sum of a local linear trend and a dummy-seasonal block (Petrís et al., 2009),

$$G = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \oplus \begin{pmatrix} -1 & -1 & -1 & \dots & -1 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix} \in \mathbb{R}^{(s+1) \times (s+1)}$$

Lastly, the innovation covariance is taken as

$$Q = \text{diag}(Q_\alpha, Q_\beta, Q_\gamma, 0, \dots, 0), \quad (6)$$

so that only the chosen coordinates receive stochastic innovations. In component form this corresponds to the familiar structural evolution equations (Harvey, 1990):

$$\alpha_t = \alpha_{t-1} + \beta_{t-1} + \omega_{\alpha,t}, \quad \omega_{\alpha,t} \sim \mathcal{N}(0, Q_\alpha), \quad (7)$$

$$\beta_t = \beta_{t-1} + \omega_{\beta,t}, \quad \omega_{\beta,t} \sim \mathcal{N}(0, Q_\beta), \quad (8)$$

$$\gamma_t = - \sum_{j=1}^{s-1} \gamma_{t-j} + \omega_{\gamma,t}, \quad \omega_{\gamma,t} \sim \mathcal{N}(0, Q_\gamma), \quad (9)$$

$$\gamma_{t-1} = \gamma_{t-2}$$

$$\vdots$$

$$\gamma_{t-s+2} = \gamma_{t-s+1}$$

with mutually independent innovations. The innovation variances $Q_\alpha, Q_\beta, Q_\gamma$ act as smoothness parameters: small values imply slowly varying components, while large values allow rapid changes. In the limiting case $Q_\alpha = Q_\beta = Q_\gamma = 0$, the structural model collapses to a deterministic linear trend with fixed seasonal pattern,

$$\alpha_t = \alpha_0 + t\beta_0, \quad (10)$$

$$\beta_t = \beta_0, \quad (11)$$

$$\gamma_t = - \sum_{j=1}^{s-1} \gamma_{t-j}, \quad (12)$$

which corresponds to classical time-linear regression with seasonal dummies (that satisfy the sum-to-zero constraint). This connects directly to the linear models discussed in the introduction: they arise when the structural components are treated as fixed coefficients rather than evolving states. When the observation density $p(\cdot)$ is Gaussian, (3) reduces to the linear-Gaussian Dynamic Linear Model (DLM) of [West and Harrison \(2006\)](#), and we reserve the term DLM for this case. For extremes we follow [Gaetan and Grigoletto \(2004\)](#) and refer to the resulting state-space model with a GEV observation layer as a Dynamic GEV (DGEV). Their work largely viewed latent evolution as a smoother for block maxima: the innovation variances were treated as tuning constants controlling roughness, selected by re-running the analysis over a small grid and judging fit through posterior predictive checks (Section 6), rather than learned jointly under a prior. [Huerta and Sansó \(2007\)](#) place DGEV modelling explicitly in an MCMC framework, combining FFBS-type latent-state simulation with prior learning of the evolution variances. They achieve conditional DLM structure by introducing an auxiliary Gaussian layer for the time-varying location. This is computationally attractive, but it adds an extra variance scale at the state level, so that multiple variance components must be inferred from the same sparse extreme-value information; these components are often weakly identified and lack a clear interpretation. Moreover, the inverse-gamma priors that are standard in much of the DLM literature ([Petris et al., 2009](#); [West and Harrison, 2006](#); [Durbin and Koopman, 2012](#); [Frühwirth-Schnatter, 1994](#)) behave poorly in the small-variance regime relevant for slowly evolving climates: they are not shrinkage priors in the sense of [Bitto and Frühwirth-Schnatter \(2019\)](#) and can be highly sensitive to hyperparameter choices ([Gelman, 2006](#); [Frühwirth-Schnatter and Knaus, 2021b](#)). In our setting this is compounded by strong posterior dependence between latent trajectories and innovation scales. We therefore adopt a reparametrisation below and introduce shrinkage priors for the innovation scales in Section 3.2; details are given in Appendix D.2.1.

2.1 Non-centred parametrisation of the structural model

For climatic applications the structural components are typically slowly varying, so the corresponding innovation scales Q_k are small (compared to observational noise). In the centred parametrisation this implies near-deterministic state increments and creates a “funnel” geometry: the latent path $\mathbf{x}_{0:T}$ and the innovation scales Q_k become strongly coupled a posteriori, which leads to very poor mixing ([Papaspiliopoulos et al., 2007](#)). We therefore adopt the non-centred parametrisation of [Frühwirth-Schnatter and Wagner \(2010\)](#) who enlarged the state of the structural DLM with one dimension (an integrated slope term), which can be viewed as an ancillary augmentation ([Yu and Meng, 2011](#)): the latent state prior is constructed to be independent of Q , so that information about Q is learned solely through the observation layer.

Let $\tilde{\mathbf{x}}_t \in \mathbb{R}^{s+2}$ denote that enlarged non-centred state. Its evolution is a linear Gaussian Markov chain with unit-variance innovations

$$\tilde{\mathbf{x}}_t = \tilde{G} \tilde{\mathbf{x}}_{t-1} + \tilde{\boldsymbol{\omega}}_t, \quad \tilde{\boldsymbol{\omega}}_t \sim \mathcal{N}(\mathbf{0}, \tilde{Q}). \quad (13)$$

where $\tilde{Q} = \text{diag}(1, 1, 0, I_{s-1})$.

All dependence on the innovation scales $Q = (Q_\alpha, Q_\beta, Q_\gamma)$ is moved from the transition equation into the location mapping,

$$\mu_t = c_t + \tilde{H}(Q) \tilde{\mathbf{x}}_t, \quad (14)$$

where c_t collects the deterministic baseline

$$c_t = \alpha_0 + t\beta_0 + \gamma_{0,t}, \quad (15)$$

and $\tilde{H}(Q)$ scales the non-centred deviations by $(\sqrt{Q_\alpha}, \sqrt{Q_\beta}, \sqrt{Q_\gamma})$. The explicit construction of $\tilde{\mathbf{x}}_t$, the decomposition into a deterministic baseline plus stochastic deviations, and the mapping back to the centred states $(\alpha_t, \beta_t, \gamma_t)$ are given in Appendix A.

3 Inference

Let

$$\boldsymbol{\theta} := (\boldsymbol{\eta}, Q, \alpha_0, \beta_0, \gamma_0) \quad (16)$$

denote the vector of static parameters, collecting the observation-layer parameters $\boldsymbol{\eta}$, (ii) the process parameters Q , and (iii) the initial-state parameters $\alpha_0, \beta_0, \gamma_0$. With prior $p(\boldsymbol{\theta})$, the joint posterior is

$$p(\mathbf{x}_{0:T}, \boldsymbol{\theta} \mid y_{1:T}) \propto p(\boldsymbol{\theta}) \prod_{t=1}^T \mathcal{N}(\mathbf{x}_t \mid G\mathbf{x}_{t-1}, Q) \prod_{t=1}^T p(y_t \mid H\mathbf{x}_t, \boldsymbol{\eta}); \quad (17)$$

and in the non-centred parametrization

$$p(\tilde{\mathbf{x}}_{0:T}, \boldsymbol{\theta} \mid y_{1:T}) \propto p(\boldsymbol{\theta}) \prod_{t=1}^T \mathcal{N}(\tilde{\mathbf{x}}_t \mid \tilde{G}\tilde{\mathbf{x}}_{t-1}, \tilde{Q}) \prod_{t=1}^T p(y_t \mid c_t + \tilde{H}(Q)\tilde{\mathbf{x}}_t, \boldsymbol{\eta}); \quad (18)$$

We sample from (18) with a two-block Gibbs scheme (Carlin et al., 1992) that alternates between updating the latent trajectory and updating the static parameters, i.e.

$$\tilde{\mathbf{x}}_{0:T} \sim p(\tilde{\mathbf{x}}_{0:T} \mid y_{1:T}, \boldsymbol{\theta}), \quad (19)$$

$$\boldsymbol{\theta} \sim p(\boldsymbol{\theta} \mid y_{1:T}, \tilde{\mathbf{x}}_{0:T}). \quad (20)$$

3.1 State update

In the DLM, the full conditional $p(\tilde{\mathbf{x}}_{0:T} \mid y_{1:T}, \boldsymbol{\theta})$ is multivariate Gaussian and we draw $\tilde{\mathbf{x}}_{0:T}$ exactly using the forward-filtering backward-sampling (FFBS) algorithm (Carter and Kohn, 1994); details are given in Appendix B. The DGEV is non-Gaussian and nonlinear in the observations; however, it retains a latent Gaussian state transition which allows us to adopt an approximate FFBS scheme based on local Laplace approximations of the GEV log-likelihood in the location μ_t (Appendix C). This is

a fast analytic alternative to more elaborate, computationally intensive sequential Monte Carlo methods (Andrieu et al., 2010; Lindsten and Schön, 2012) and is expected to be accurate when the log-likelihood is close to quadratic in the location over the posterior-relevant region.

3.2 Parameter update

We next specify prior distributions for all static parameters and describe the corresponding update steps. All full conditional distributions required are stated in Appendix D.

Observation layer.

In the DLM we place the standard inverse-gamma prior on the Gaussian observation variance,

$$\sigma^2 \sim \text{IG}(a_\sigma, b_\sigma),$$

which yields a conjugate Gibbs update (West and Harrison, 2006). In the DGEV we use the same prior for the GEV scale via $\sigma^2 \sim \text{IG}(a_\sigma, b_\sigma)$ and assign a bounded uniform prior to the GEV shape parameter,

$$\xi \sim \mathcal{U}(\xi_{\min}, \xi_{\max}),$$

with $\xi_{\min} = -0.5$ and $\xi_{\max} = 0.5$. This is consistent with empirical guidance in climate and hydrology where ξ is often modest (between -0.3 and 0.3, Katz et al. (2002)). The observation parameters (σ, ξ) do not admit closed-form conditionals and are therefore updated by a random-walk Metropolis–Hastings step.

State layer.

A central modelling decision concerns the innovation variances. As discussed in 2.1, slowly varying environmental signals typically imply very small Q_k ; in this regime inverse-gamma priors can over-regularise or, conversely, place too much mass away from zero, leading to undesirable behaviour (Gelman, 2006; Frühwirth-Schnatter and Wagner, 2010). We parameterise $Q_k = q_k^2$ in terms of innovation standard deviations q_k and impose a global–local shrinkage prior that concentrates mass near zero while retaining heavy tails. Specifically, for each dynamic block $k \in \{\alpha, \beta, \gamma\}$ we use the hierarchical Bayesian lasso prior (Park and Casella, 2008; Belmonte et al., 2014)

$$q_k \mid \tau_k \sim \mathcal{N}(0, \tau_k), \quad \tau_k \mid \lambda^2 \sim \text{Exp}\left(\frac{\lambda^2}{2}\right), \quad \lambda^2 \sim \text{Gamma}(a_\lambda, b_\lambda), \quad (21)$$

with a common global shrinkage parameter λ and local scales τ_k . For the deterministic baselines in (15), we assign Gaussian priors

$$\alpha_0 \sim \mathcal{N}(m_\alpha, P_\alpha), \quad \beta_0 \sim \mathcal{N}(m_\beta, P_\beta), \quad \gamma_0 \sim \mathcal{N}(\mathbf{m}_\gamma, P_\gamma), \quad (22)$$

centered at zero with $P_\alpha, P_\beta, P_\gamma$ diffuse (~ 100).

The choice of these priors is very natural - at least if we work with the pseudo-Gaussian approximation for extremes. Conditional on the auxiliary scales τ_k , this yields a joint conjugate multivariate Normal update for $(\alpha_0, \beta_0, \gamma_0, q_\alpha, q_\beta, q_\gamma)$, while (τ_k, λ) are updated from their standard conditionally conjugate full conditionals.

Finally, because the likelihood depends on the innovation scales only through q_k^2 , the posterior is invariant under simultaneous sign changes of q_k and the associated non-centred state block. Following [Frühwirth-Schnatter and Wagner \(2010\)](#), we apply random sign switches after each iteration to improve mixing.

4 Time-varying exceedance risk and return periods

In a nonstationary climate, return levels and return periods must be interpreted with care: because our structural DGEV model makes the block-maxima distribution time dependent through latent states, we follow [Katz et al. \(2002\)](#); [Katz \(2013\)](#); [Cheng et al. \(2014\)](#) and treat them as instantaneous (or effective) risk measures, conditional on the climate state at time t (as if the climate were frozen at that state).¹

We focus here on time-varying exceedance risk. Fix a high threshold y^* . The conditional exceedance probability at time t is

$$p_t(y^*) = \Pr(Y_t > y^* \mid \mathbf{x}_t, \boldsymbol{\theta}) = 1 - G_t(y^*),$$

and the corresponding effective return period is

$$R_t(y^*) = \frac{1}{p_t(y^*)}. \quad (23)$$

Bayesian inference yields a full posterior distribution over the entire trajectory $\{p_t(y^*)\}_{t=1}^T$ (and hence $\{R_t(y^*)\}_{t=1}^T$): for posterior draws $\{(\mathbf{x}_{0:T}^{(m)}, \boldsymbol{\theta}^{(m)})\}_{m=1}^M$, we evaluate $G_t(\cdot)$ at the implied $(\mu_t^{(m)}, \sigma^{(m)}, \xi^{(m)})$ and compute $p_t^{(m)}(y^*) = 1 - G_t^{(m)}(y^*)$.

To express risk on an annual scale (exceedance possible in every season), suppose there are s blocks per calendar year. Let S_j be the set of block indices in year j and define the annual maximum $M_j := \max_{t \in S_j} Y_t$. Conditionally on $(\mathbf{x}_{0:T}, \boldsymbol{\theta})$, block maxima are independent, hence

$$\Pr(M_j \leq y^* \mid \mathbf{x}_{0:T}, \boldsymbol{\theta}) = \prod_{t \in S_j} \Pr(Y_t \leq y^* \mid \mathbf{x}_t, \boldsymbol{\theta}) = \prod_{t \in S_j} G_t(y^*). \quad (24)$$

Therefore, the conditional probability of at least one exceedance of y^* in year j is

$$p_j(y^*) = \Pr(M_j > y^* \mid \mathbf{x}_{0:T}, \boldsymbol{\theta}) = 1 - \prod_{t \in S_j} G_t(y^*), \quad (25)$$

¹See also [Salas and Obeysekera \(2014\)](#); [Cooley \(2012\)](#) for discussion of return level definitions under nonstationarity.

and the associated annual return period is

$$R_j^{(\text{ann})}(y^*) = \frac{1}{p_j(y^*)}. \quad (26)$$

5 Forecasting

Forecasts follow directly from the generative state-space model (2)–(3). Given posterior draws $\{(\mathbf{x}_{0:T}^{(m)}, \boldsymbol{\theta}^{(m)})\}_{m=1}^M$ and horizon h (months), we obtain a posterior predictive sample by forward simulation: for $\ell = 1, \dots, h$,

$$\mathbf{x}_{T+\ell}^{(m)} \mid \mathbf{x}_{T+\ell-1}^{(m)}, \boldsymbol{\theta}^{(m)} \sim \mathcal{N}(G\mathbf{x}_{T+\ell-1}^{(m)}, Q^{(m)}), \quad (27)$$

$$Y_{T+\ell}^{(m)} \mid \mathbf{x}_{T+\ell}^{(m)}, \boldsymbol{\theta}^{(m)} \sim p(\cdot \mid \mu_{T+\ell}^{(m)}, \boldsymbol{\eta}^{(m)}), \quad \mu_{T+\ell}^{(m)} = H\mathbf{x}_{T+\ell}^{(m)}. \quad (28)$$

If we want to report on an annual scale, we can simply coarse-grain the predictive draws without having to refit an annual model. Using the same notation as in section 4 we apply the aggregation operators (averaging and maximizing) again to the fine-scale draws. For averages,

$$\bar{Y}_j^{(m)} = \frac{1}{|S_j|} \sum_{t \in S_j} Y_t^{(m)}$$

and for extremes,

$$M_j^{(m)} = \max_{t \in S_j} Y_t^{(m)}.$$

6 Model diagnostics

The probability integral transforms (PITs)

$$u_t = \mathbb{P}(Y_t < y_t \mid \mathbf{x}_t, \boldsymbol{\eta}) = \int_{-\infty}^{y_t} p(y \mid \mathbf{x}_t, \boldsymbol{\eta}) \, dy, \quad t = 1, \dots, T,$$

should behave approximately as i.i.d. $U(0, 1)$ under correct specification (Rosenblatt, 1952). Since $\mathbf{x}_t$ and $\boldsymbol{\eta}$ are unknown, we propagate this uncertainty by computing PITs per posterior draw: for samples $\{(\boldsymbol{\eta}^{(m)}, \mathbf{x}_{1:T}^{(m)})\}_{m=1}^M$,

$$u_t^{(m)} = \mathbb{P}(Y_t < y_t \mid \mathbf{x}_t^{(m)}, \boldsymbol{\eta}^{(m)}), \quad t = 1, \dots, T,$$

which yields a posterior sample $\{u_t^{(m)}\}_{m=1}^M$ for each u_t . This provides us with a visual tool to assess the uniformity of the PITs, either directly on a histogram or via a PP plot. Following Gaetan and Grigoletto (2004), we can quantitatively check goodness-of-fit with a posterior predictive calibration procedure based on the Kolmogorov-Smirnov test (Gelman et al., 1996). For each draw m , we compute the observed discrepancy

$$\text{KS}_{\text{obs}}^{(m)} = \text{KS}(u_{1:T}^{(m)}, U(0, 1)),$$

simulate a replicate series $y_{1:T}^{\text{rep},(m)} \sim p(\cdot \mid \boldsymbol{\eta}^{(m)}, \mathbf{x}_{1:T}^{(m)})$, and form replicate PITs and KS test statistics

$$u_t^{\text{rep},(m)} = \mathbb{P}(Y_t < y_t^{\text{rep},(m)} \mid \mathbf{x}_t^{(m)}, \boldsymbol{\eta}^{(m)}), \quad \text{KS}_{\text{rep}}^{(m)} = \text{KS}\left(u_{1:T}^{\text{rep},(m)}, U(0, 1)\right).$$

The (Bayesian) posterior predictive p -value is then

$$p_B = \frac{1}{M} \sum_{m=1}^M \mathbb{I}\left\{\text{KS}_{\text{rep}}^{(m)} \geq \text{KS}_{\text{obs}}^{(m)}\right\}.$$

After the in-sample checks, we assess out-of-sample predictive performance. We use an expanding-window cross-validation (CV) scheme: for a split time t_s we refit the model on the training prefix $y_{1:t_s}$, generate posterior predictive draws for the next H months as in Section 5, and compare these forecasts to the held-out observations $y_{t_s+1:t_s+H}$.

For each split t_s we report the coverage

$$\text{Cov}_{1-\alpha}(t_s) = \frac{1}{H} \sum_{h=1}^H \mathbb{I}\left\{y_{t_s+h} \in [\hat{q}_{\alpha/2}(t_s+h), \hat{q}_{1-\alpha/2}(t_s+h)]\right\},$$

where α is the significance level and $\hat{q}$ a predictive quantile, and the point accuracy

$$\text{RMSE}(t_s) = \left(\frac{1}{H} \sum_{h=1}^H (\hat{q}_{0.50}(t_s+h) - y_{t_s+h})^2 \right)^{1/2},$$

summarised by the root mean square error wrt the predictive median.

7 Data

The data in this study are provided by the Royal Meteorological Institute of Belgium and consist of daily observations of maximum temperature (TX) and minimum temperature (TN) at Uccle from 1892 to 2022. To connect our structural time-series framework with classical asymptotic theory, we construct monthly summaries. For temperature extremes we use the standard indices ([Expert Team on Climate Change Detection and Indices \(ETCCDI\), 2024](#)): TXx and TXn denote, respectively, the monthly maximum and minimum of daily maximum temperature, while TNx and TNn are the monthly maximum and minimum of daily minimum temperature. Monthly averages of TX and TN (denoted TXm and TNm) serve as inputs to the Gaussian DLM for the bulk component, for which a central-limit argument motivates approximate normality at the monthly scale. The extreme-value component is driven by the block extremes TXx, TXn, TNx and TNn, for which the Fisher–Tippett–Gnedenko theorem motivates a GEV observation model ([Fisher and Tippett, 1928](#)).

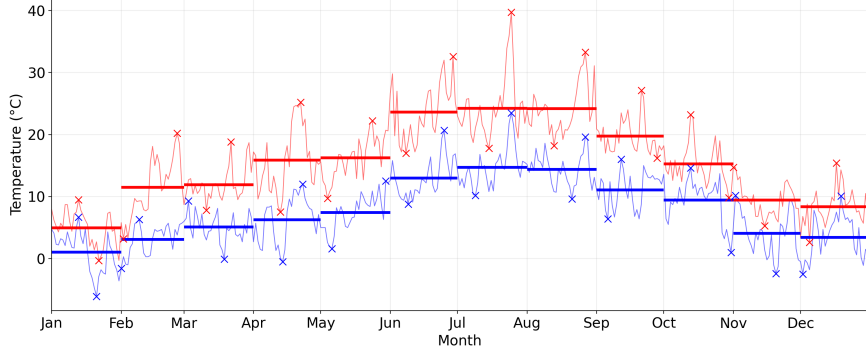


Fig. 2: Daily maximum temperature TX (red) and daily minimum temperature TN (blue) in Uccle during the record year 2019 (Vautard et al., 2020), together with the six monthly temperature summaries used in this paper. Horizontal lines show monthly means (TXm in red, TNm in blue). Crosses mark the within-month extremes: for TX, red crosses indicate the monthly maximum (TXx) and monthly minimum (TXn) of daily TX; for TN, blue crosses indicate the monthly maximum (TNx) and monthly minimum (TNn) of daily TN.

8 Results

The models were fitted on the above 6 monthly temperature summaries in a PYTHON environment on a Linux system (2 cores on an Intel Xeon Gold 6240 processor, 6.6 GB RAM). One MCMC run (20 000 iterations; 5 000 burn-in; no thinning) took approximately 3 hours. We start by characterising changes in the bulk of the temperature distribution using the structural DLM fitted to TXm and TNm. Figure 3 shows posterior medians of the latent level α_t ($^{\circ}\text{C}$) and slope β_t ($^{\circ}\text{C}/\text{decade}$), with 90% credible intervals. The static parameters σ, Q indicate that the extracted mean signals in Figure 3 are indeed smooth relative to the residual month-to-month variability (that is, $\sigma \gg \sqrt{Q}$, see Figure F2). For TXm, the observation noise is $\sigma = 1.86$ ([1.80, 1.92]), and for TNm $\sigma = 1.62$ ([1.57, 1.67]).

For the mean nighttime temperatures TNm there has been a consistent acceleration. The level rises from about 5.4°C in 1892 to 7.7°C in 2020, and the slope increased notably during the 1950–1980 period, indicating a higher rate of warming toward the end of the record. For the mean daytime temperatures TXm, the level also increased substantially, from about 12.4°C in 1892 to 15.4°C in 2020, but the slope pattern is more stable. In particular, after the 1950–1980 plateau/level shift there is limited evidence that the warming rate itself has accelerated. Overall, the structural DLM analysis suggests that changes in mean temperature are more pronounced for night-time than for day-time temperatures: TNm appears to have undergone a catch-up phase via its mid-century slope increase, and at the end of the record it warms approximately at the same rate as TXm.

We now turn to the extreme-value component, fitting the structural DGEV model to the monthly temperature indices TXx, TXn, TNx and TNn. Figure 4 shows again

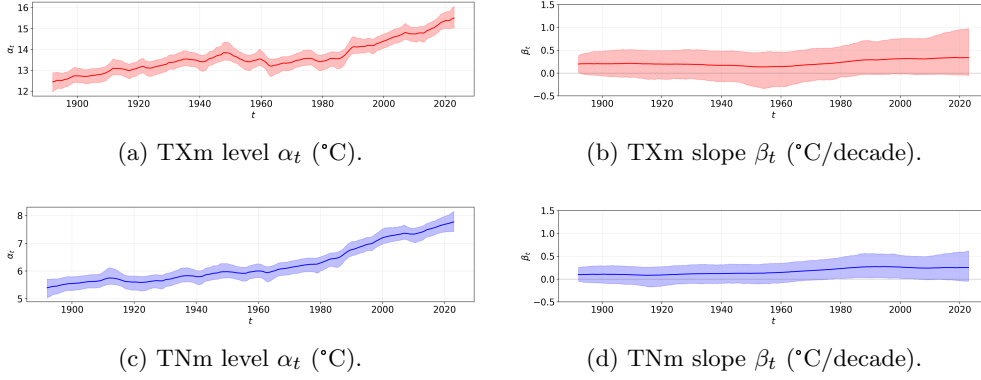


Fig. 3: Structural DLM decomposition of mean temperatures in Uccle for monthly mean maxima (TXm, top row) and monthly mean minima (TNm, bottom row). The level α_t represents the annual mean temperature ($^{\circ}\text{C}$), and the slope β_t represents the local rate of change ($^{\circ}\text{C}$ per decade). Solid lines denote posterior medians and shaded bands denote 90% credible intervals.

state evolution for α_t and β_t . The posterior histograms of the static parameters are displayed in Figure F3. Across all four indices, the posterior for the extreme value index ξ is negative, indicating a Weibull-type tail (finite upper endpoint, see Appendix E). Posterior medians (90% credible intervals) for the tail parameters are: TXx $\xi = -0.28$ ($[-0.30, -0.26]$), $\sigma = 2.63$ ($[2.55, 2.72]$); TXn $\xi = -0.14$ ($[-0.16, -0.12]$), $\sigma = 2.27$ ($[2.19, 2.34]$); TNx $\xi = -0.28$ ($[-0.30, -0.26]$), $\sigma = 1.92$ ($[1.86, 1.99]$); TNn $\xi = -0.05$ ($[-0.07, -0.02]$), $\sigma = 2.12$ ($[2.05, 2.20]$). Notably, the coldest day of the month TNn has the “heaviest” tail among the four, with ξ closest to zero.

The clearest dynamical feature concerns the warmest nights (TNx). The slope increases markedly in the 70s as can be seen in Figure 4 (e) and (f),: $\beta_{1892} = 0.05$ ($[-0.40, 0.40]$) and it rises to $\beta_{2020} = 0.27$ ($[-0.14, 0.75]$). The highest rate is unsurprisingly obtained by the hottest days: the slope of TXx increases toward the end of the record with $\beta_{2020} = 0.56$ ($[-0.18, 1.49]$), though with considerable higher uncertainty compared to the other indices. We also have a clear signal of mid-century dampening as we saw in the DLM analysis, especially for TXx. TXn (the temperature of the coldest day) exhibits a similar but weaker dampening. Finally, for the cold-night minima TNn we do not see such a plateau: the level increased steadily and the slope increased only moderately (yet uncertain), especially compared to TNx. The fit for TNn however is unsatisfactory (see 8.3) so we abstain from overinterpreting this result.

8.1 Return periods

In July 2019, Uccle set a new 82-year record for TXx. The previous record was 36.8°C (June 1947), which was surpassed by no less than 2.9°C . Figure 5 puts this jump into probabilistic context by tracking the return period of both records. By the end of the sample, exceeding the 1947 record has a return period of roughly a decade, which helps explain why only three years after 2019 the record was broken again (to 38.1°C).

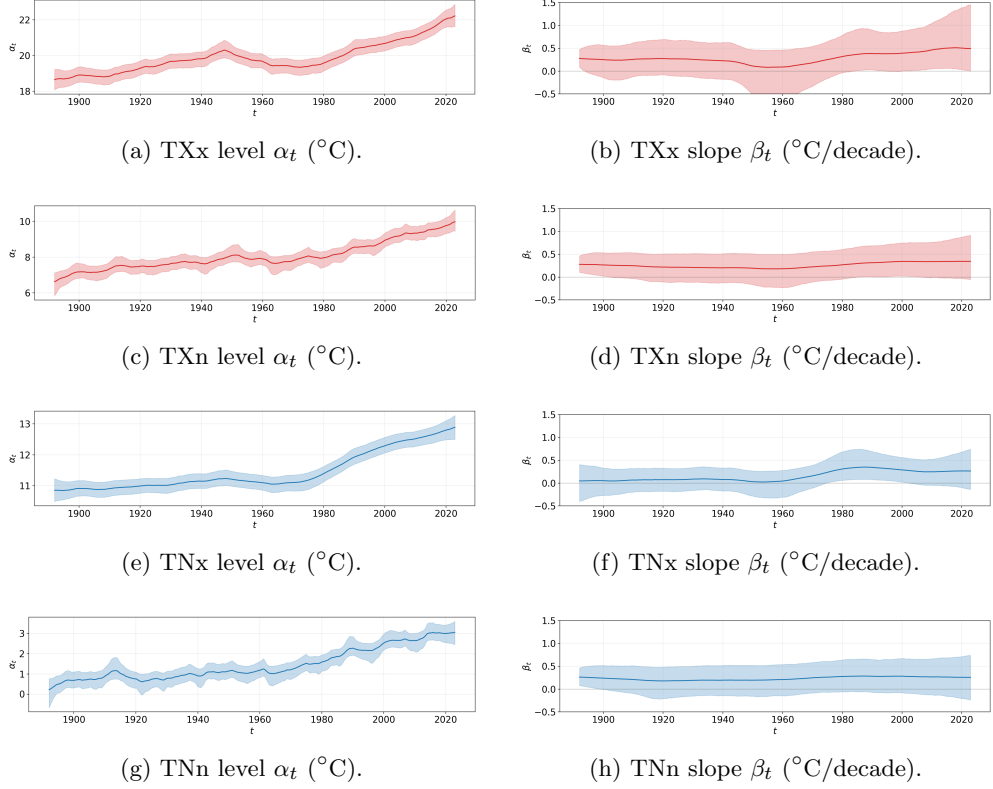


Fig. 4: Structural DGEV decomposition of monthly temperature extremes in Uccle. Rows correspond (top to bottom) to TXx, TXn, TNx, and TNn. The left column shows posterior medians of the latent level α_t (in $^{\circ}\text{C}$), and the right column shows posterior medians of the latent slope β_t (in $^{\circ}\text{C}/\text{decade}$). Shaded bands denote 90% credible intervals.

In contrast, values near 40°C still have extremely long return periods (~ 1000 years), similar to the kind of event 36.8°C was 50 years ago. It is instructive to see what happened in 2010: a previously practically impossible (return period > 10000 years) event suddenly emerges with rapidly increasing probability. Because the TXx tail is strongly bounded ($\xi < 0$), the model implies a finite upper limit (the 100% quantile) for each year. One can derive (appendix E) that when the endpoint crossed 39.7°C the risk inflated rapidly with a power of $1/|\xi| \approx 3.5$ out of nothingness. The right panel of Figure 5 visualizes that TXx endpoint over time, which, as a quantile, changes parallel with the shifting location. Temperatures close to the 2019 record indeed only become assigned non-negligible probability from the early 2010s onward, when the endpoint transitions past that range.

Similarly we quantify warm-night risk here. A widely used indicator is the number of tropical nights (TR; here implemented as the probability that $\text{TNx} > 20^{\circ}\text{C}$ for

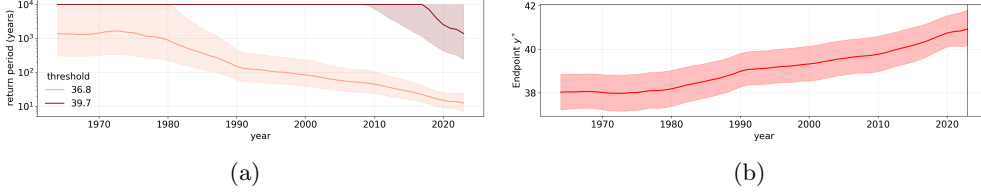


Fig. 5: TXx risk metrics over time. Left: posterior return periods (values above 10^4 years clipped for readability) for exceedances of the 1947 and 2019 records. Color code is on the figure. Right: trajectory of the endpoint.

Table 1: Posterior predictive annual summaries (median, 5% quantile, 95% quantile) for the Uccle temperature record.

Year	TXm	TNm	TXx	TXn	TNx	TNn
2022	15.6 ^{16.6} _{14.5}	7.8 ^{8.7} _{7.0}	34.1 ^{37.5} _{31.0}	-1.1 ^{1.8} _{-5.2}	20.6 ^{23.0} _{18.4}	-6.2 ^{-3.2} _{-11.0}
2032	15.9 ^{17.5} _{14.4}	8.0 ^{9.1} _{7.0}	34.6 ^{38.2} _{31.2}	-0.8 ^{2.4} _{-5.0}	20.9 ^{23.4} _{18.5}	-6.0 ^{-3.0} _{-10.9}
2042	16.2 ^{18.6} _{14.3}	8.3 ^{9.8} _{6.9}	35.1 ^{39.3} _{31.4}	-0.4 ^{3.0} _{-4.8}	21.2 ^{23.9} _{18.6}	-5.7 ^{-2.1} _{-10.9}

at least one day; Appendix Figure F1). Comparing 1900, 1970 and 2020, this probability of a TR increases from 0.144 [0.097, 0.202] to 0.190 [0.136, 0.252] and then to 0.626 [0.531, 0.717], implying that tropical nights have in recent decades shifted from occasional events to almost certain events (return period 1-2 years) in Belgium.

8.2 Forecasts

We forecast 20 years ahead ($h = 240$ months) and coarse-grain to an annual scale for reporting as explained in Section 5. Table 1 summarises three reference years: 2022 (end of the observation record), 2032 (+10 years) and 2042 (+20 years).

Figure F6 shows posterior predictive fan charts for TXm and TNm, together with the corresponding annual-mean forecasts: the predictive median for TXm increases by about 0.65°C from 2022 to 2042, while TNm increases by 0.50°C ; with uncertainty naturally increasing with forecast horizon due to uncertainty in future trends as well as irreducible observation noise.

Figure F7 reports the corresponding forecasts under the structural DGEV model for the four monthly extreme-temperature indices. Among the warm extremes, TXx shows the strongest increase in the predictive median (about $+1^\circ\text{C}$ from 2022 to 2042).

For cold-day extremes, TXn shifts upward toward 0°C at a rate of about $+0.35^\circ$ per decade. In 30 years, if the current trend persists, we can expect the first years without "ice days" (ID; Expert Team on Climate Change Detection and Indices (ETCCDI) (2024)). Finally, night-time extremes evolve more moderately in the medians: TNx increases by about $+0.50^\circ\text{C}$ from 2022 to 2042, while TNn increases by about $+0.38^\circ\text{C}$.

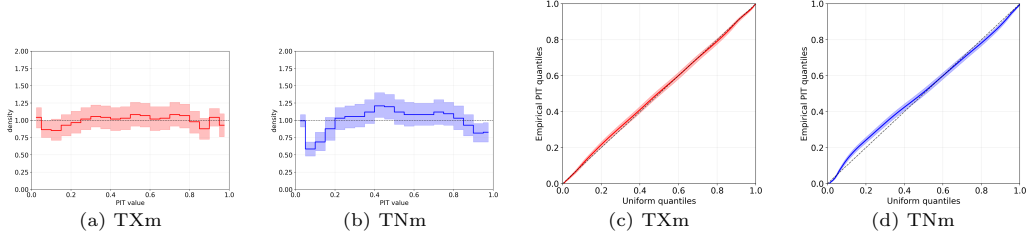


Fig. 6: Goodness-of-fit diagnostics for the Gaussian DLM fitted to monthly mean temperatures. (a) and (b) : PIT histograms (uniformity check) for TXm and TNm. (c) and (d): PIT PP plots comparing empirical PIT quantiles to $U(0,1)$.

8.3 Goodness-of-fit

As explained in section 6 we describe goodness-of-fit via PIT histograms (and a corresponding uniformity test) and PP plots. For the averages, this is summarised in Figure 6. For TXm we obtain a p-value $p_B = 0.316$, and both the PIT histogram and PP plot are satisfactory. For TNm, however, the p-value is significant ($p_B = 0.0078$) and the corresponding panels in Figure 6 show systematic deviations from uniformity, indicating that the fitted structural DLM does not fully capture the distributional behaviour of monthly mean night-time temperatures. For extremes, Figure 7 reports the same diagnostics for TXx, TXn, TNx and TNn. The fits for TXx and TNx are satisfactory with $p_B = 0.17$ and $p_B = 0.21$, respectively, and the PP plots track the uniform diagonal closely. The cold indices show an inadequate fit: TXn yields $p_B = 0.07$ with visible departures in the PIT-based diagnostics, similarly TNn exhibits pronounced non-uniform PITS ($p_B = 0.009$). These results suggest that the current specification may be slightly over-simplified at least for TNm, TXn and TNn; we return to this point in the discussion. Now we move on to out-of-sample performance.

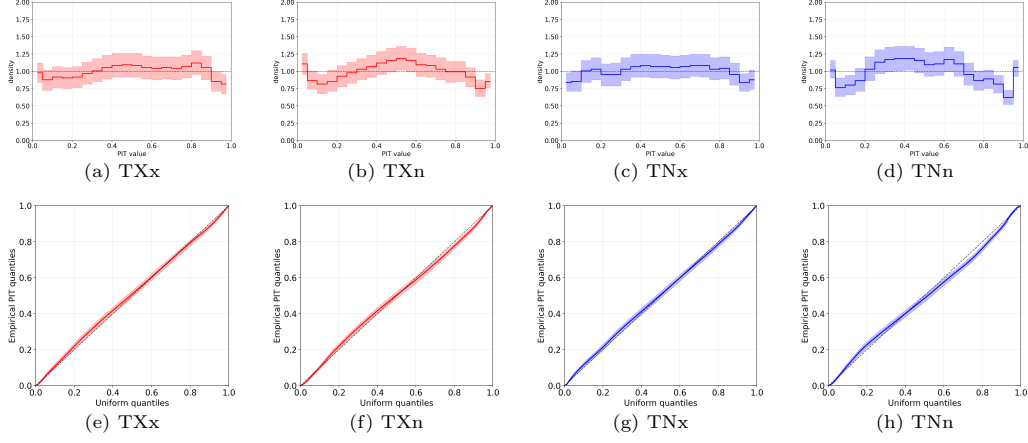


Fig. 7: Goodness-of-fit diagnostics for the structural DGEV fitted to monthly temperature extremes. Top row: PIT histograms. Bottom row: PIT PP plots against $U(0, 1)$. Columns (left to right): TXx, TXn, TNx, TNn.

Following the CV scheme in section 6 we refit the DLM on training prefixes ending at t_s and generate draws for the next $H = 120$ months to check our predictive performance. We consider split points at 60%, 80%, and 90% of the record (approximately 1970, 1997, and 2010 for the Uccle series). Figure F8 shows the resulting 10-year-ahead forecast fans for TXm and TNm. Table 2 reports coverage and RMSE. Coverage remains close to the nominal level across splits, and TNm forecasts are more accurate (lower RMSE) than TXm, consistent with TNm exhibiting less intrinsic variability around its latent mean ($\sigma \approx 1.86$ for TXm and $\sigma \approx 1.62$ for TNm).

Table 2: Rolling-origin CV metrics for mean temperatures (horizon $H = 10$ yrs, nominal level 0.9): empirical coverage of the 90% predictive interval and RMSE of the predictive median.

Split point	TXm		TNm	
	Coverage	RMSE	Coverage	RMSE
0.6	0.9250	1.7568	0.9583	1.3965
0.8	0.9583	1.6888	0.9083	1.5164
0.9	0.8917	2.0093	0.9417	1.6160

For the day-and night time extremes, figures F9 and F10 show the 10-year-ahead forecasts while Table 3 summarises coverage and predictive accuracy.

Table 3: Rolling-origin CV metrics for extreme-temperature indices (horizon $H = 10$ yrs, nominal level 0.9).

Split point	TXx		TXn		TNx		TNn	
	Coverage	RMSE	Coverage	RMSE	Coverage	RMSE	Coverage	RMSE
0.6	0.9500	2.4744	0.8750	2.3869	0.9333	1.7999	0.9333	2.4277
0.8	0.9583	2.2984	0.8833	2.5214	0.9500	1.7502	0.8500	2.6313
0.9	0.9000	2.8077	0.9083	2.4987	0.8833	2.0142	0.9167	2.4610

9 Discussion

We summarized the Uccle temperature record into six complementary monthly series—two mean metrics and four extreme metrics—so that changes in the bulk and in the tails can be compared within one modelling framework. For the mean series, the seasonal geometry was well identified yet essentially unchanged across the record: the patterns at early/mid/late reference times nearly coincide and their uncertainty bands overlap strongly (Appendix F.2, Fig. F4). Non-stationarity is driven by a gradual slope increase, for average night (TNm) and day (TXm) temperatures and, specifically for TXm by a mid-century plateau, that is likely related to the aerosol effect (Wild, 2009). The structural DLM implied an increase of roughly 3°C in TXm and about 2.3°C in night temperatures TNm from 1892 to 2022; with evidence of late-century (1960-1980) acceleration for TNm. By the end of the record, the inferred warming-rate (slope) is about 0.3°C per decade for both TXm and TNm. We then turned to tail-specific changes.

For the extreme indices, the structural DGEV yielded an analogous decomposition, with largely time-invariant seasonality (although a small, non-significant redistribution of seasonal amplitudes might be argued for TNx; see Fig. F5) but substantial evolution in level and slope. Among the extremes, TNx exhibited the strongest and fastest acceleration during the 1960–1980 period, after which the warming rate stabilized at about a $5\times$ higher slope ($\sim 0.25^{\circ}\text{C}$). The TXx trend steepened more gradually toward the end of the record: the current slope for the expected hottest day is about 0.5°C per decade (though with considerable uncertainty; see Fig. F3). Posterior predictive simulations further suggest that “ice-day” risk is shrinking: within a few decades the expected annual TXn rises above 0°C (Figure F7), so if the expected coldest daytime maximum is above freezing, we may no longer reasonably expect white Christmases. We also examined another common climate index, “tropical nights” ($\text{TNx} > 20^{\circ}\text{C}$): over the last 50 years, the probability of at least one tropical night in a given year tripled (from 19% to 63%).

The tropical-night result provided a concrete illustration of risk tracking with the structural DGEV. Once the model is fitted, exceedance probabilities for any given threshold—and their evolution through time—are directly available. This avoids the statistically doubtful practice of fitting separate models over moving windows (Kharin and Zwiers, 2000, 2005; Kharin et al., 2007, 2013) and yields a full posterior over risk trajectories. A further practical benefit is that, once the static parameters

are learned (from a sufficiently long record), the model can be deployed online: as new observations arrive they can be assimilated sequentially via filtering, hence updating risk and forecasts in real time. In this way, persistent departures from the current predictive distribution (i.e. repeated outcomes above or below expectation given the current level, slope and seasonal amplitudes) translate into posterior evidence for changes in the underlying state, which allows emerging accelerations or decelerations in hazard to be detected as part of routine monitoring; a useful perspective given how challenging formal detection of possible trend changes (using discrete changepoints) can be (Beaulieu et al., 2024; Cahill et al., 2015).

As a societally urgent example, we applied the same risk tracking to TXx, the hottest day of the year in Uccle, and summarized its evolution throughout the record via the time-varying hazard curves (Fig. 5, left). These curves place recent record-breaking heat in a probabilistic context: the near-40°C event in 2019 shattered an 82-year record by almost 3°C, yet under the current fitted climate state breaking the pre-2019 record is already expected to occur at least once per decade (coming from a 1000-year event). In contrast, exceeding the 2019 record itself remains an extreme tail event (outlier) at present, with a return period on the order of 10^3 years, although the model naturally implies this return period will shorten as the latent climate state continues to drift. These disproportionate (non-linear) changes in far-tail risk under modest (post-1970 approximately linear) warming are well known (Rahmstorf and Coumou, 2011; Coumou et al., 2013). Robinson et al. (2021) documents the emergence of extremely rare 4σ heat anomalies all over the globe that are essentially absent in the mid-20th-century reference climate and become detectable only now - in an anthropogenically altered climate. However, much of this intuition is phrased in Gaussian language (“ σ ” events), whereas temperature (and definitely not maximum temperature) is not well described by a Gaussian (tail). Our DGEV inference concentrated strongly on a negative shape for the warm-maxima series, which implies a bounded (Weibull-type) upper tail; such an effective upper bound on extreme temperatures has recently also been argued for on physical grounds (Zhang and Boos, 2023; Nicolas and Hotz, 2025). Reaching 40 degrees in Uccle was, theoretically, simply impossible until ~ 15 years ago. Under evolution of bounded tails (Fig. 5, right) there is an emergence of these previously impossible events built in: as the effective upper endpoint approaches a high threshold, exceedance probabilities can move from literal zero to non-negligible values over a short time window (or, equivalently, return periods from infinity to the course of a human life span). Appendix E formalizes this transition and shows that, for near-endpoint thresholds, the rise in exceedance risk follows to first order a steep power-law increase (with exponent roughly 3–4) over time, which helps explain the growing prominence of record-shattering heat extremes (Fischer et al., 2021a, 2025).

Out-of-sample validation was encouraging. In expanding-window cross-validation, both accuracy and precision remained fairly stable, which showed that the fitted structural signals generalize well and yield truthful predictive uncertainty at least at decadal horizons. In-sample calibration, however, reveals a clearer limitation on the cold side. Probability integral transform (PIT) diagnostics show systematic deviations for the left-tail indices, most notably TXn and TNn, indicating a lack of fit in the lower tail. For this reason we do not emphasize detailed inference for these minima:

the model is adequate for tracking broad nonstationary signals, but it is less reliable for fine-grained tail statements on the coldest extremes.

Several extensions are natural within the same modular framework (in order of importance). First, although the location-driven specifications were intuitive and already yielded clear insights, it is important to allow the scale σ_t to vary, either deterministically (seasonal/covariate-driven) or stochastically (latent volatility). This is particularly relevant for variables where nonstationarity enters primarily through variability rather than location, with precipitation as the obvious next target: rainfall extremes are fundamentally heavy-tailed and intensification is often expressed through changes in scale (and sometimes shape; Jayaweera et al. (2025)). Global analyses document widespread increases in extreme rainfall with strong spatial contrasts (e.g. Asadieh and Krakauer, 2015; Westra et al., 2013), and recent nonstationary GEV applications use time, global-mean temperature, or regional indices as covariates driving multiple parameters (Jorgensen and Nielsen-Gammon, 2024; Razi Ghalavand et al., 2025). There is also evidence for changing seasonality in precipitation extremes (Gu et al., 2017), which was absent in the temperature case. Drought is another natural application but conceptually harder: rainfall deficit alone does not imply drought (Van Loon, 2015), and interacting processes in the hydrological cycle motivate multivariate and spatial extensions.

Second, the state space can directly be augmented with additional (static or dynamic) regression coefficients (Scott and Varian, 2014), allowing the latent climate state to be partially anchored to physical drivers (NAO, ENSO or a circulation proxy as in Zhang and Boos (2023)) alongside its structural evolution.

Third, on the extremes side one can move beyond block maxima when persistence or clustering is the target. A direct extension is to model r -largest order statistics (Coles et al., 2001, Sec. 3.5). More generally, heatwaves and cold spells motivate threshold-based models that can represent runs and clusters. As noted already by Gaetan and Grigoletto (2004), a threshold formulation introduces its own difficulties (threshold choice and the resulting effective sample size), but extending the present structural approach to a POT layer (possibly with declustering) is a compelling direction for extremes-anchored time-series modelling.

Fourth, the PIT deviations for the minima underscore the need to benchmark our approximate inference before any extension can be made. A systematic comparison of the Laplace-based state update against recent particle MCMC developments (Andrieu et al., 2010; Lindsten et al., 2014) would clarify if and when the approximation is adequate and when we should use exact sequential Monte Carlo techniques for the state update.

Fifth, and most importantly as the framework grows, is model discrimination: whether nonstationarity is warranted in a given record, and in which component, is itself a time-series question—nonstationarity should not be presumed to occur everywhere simply because of climate change (Slater et al., 2020). The stability of seasonal geometry across the indices suggests that nonstationarity in temperature is typically carried by the slowly evolving level/slope rather than by a re-shaping of the annual cycle, so allowing seasonality to be time-varying should be treated as a selectable refinement rather than a default. "Turning off" a component however corresponds

to an innovation variance collapsing to zero, i.e. a boundary problem. Since classical hypothesis tests cannot be applied in this case, one can pose component inclusion via Bayesian model selection, letting component enter or exit the state through stochastic search specification (Frühwirth-Schnatter and Wagner, 2010; Cadonna et al., 2020), or through dimension-changing reversible-jump MCMC moves (Green, 1995). Either route yields posterior evidence for which structural components must be dynamic, which can remain fixed and which are identically zero (a stationary model).

10 Conclusion

To conclude, stationarity might indeed be dead for many environmental extremes, and modern risk assessment should take that seriously. Dynamic extremes turn classical return-period thinking into a time-series problem: risk is not a single number but a trajectory. We developed a modular structural extreme-value state-space framework that makes this operational, delivering time-varying exceedance probabilities (and associated return-period quantities) with coherent Bayesian uncertainty from a single observational record, and let each summary of an environmental variable carry its own nonstationary components—rather than imposing bulk dynamics on the tails (“detrending”). Methodologically, we casted the structural DGEV in a modern probabilistic state-space form with principled filtering and forecasting logic, deliberately staying as close as possible to familiar linear trend–seasonal decompositions while retaining a parametric extreme-value observation model for rare-event inference. We made Bayesian learning feasible in the slow-signal regime of long records by stabilizing MCMC in the near–zero-innovation setting via a non-centred augmented parametrisation, adopting shrinkage priors that remain well behaved when evolution variances are small, and using a fast local Laplace step for practical non-Gaussian state updates. In the bounded-temperature case, the framework further clarifies why and how record risk can accelerate sharply as previously impossible events begin to emerge. Finally, because the formulation remains modular and scientifically interpretable, the extension path is straightforward; we therefore advocate that (structural) extreme-value state-space models be explored more widely as a practical, decision-relevant approach to nonstationary hazard assessment in long environmental records.

Appendix A Non-centering of the structural state space model

In the centred parametrisation the latent structural state follows

$$\mathbf{x}_t \mid \mathbf{x}_{t-1}, Q \sim \mathcal{N}(G\mathbf{x}_{t-1}, Q), \quad \mathbf{x}_t = (\alpha_t, \beta_t, \boldsymbol{\gamma}_t^\top)^\top, \quad (\text{A1})$$

where α_t is the local level, β_t the slope, and $\boldsymbol{\gamma}_t \in \mathbb{R}^{s-1}$ the seasonal dummy state. Let $S \in \mathbb{R}^{(s-1) \times (s-1)}$ denote the dummy-rotation transition matrix, so that the seasonal block of G equals S . Only the first seasonal coordinate receives an innovation. Writing $\mathbf{e}_1 = (1, 0, \dots, 0)^\top \in \mathbb{R}^{s-1}$, we take the innovation covariance as

$$Q = \text{diag}(Q_\alpha, Q_\beta) \oplus (Q_\gamma \mathbf{e}_1 \mathbf{e}_1^\top) = \text{diag}(Q_\alpha, Q_\beta, Q_\gamma, 0, \dots, 0), \quad (\text{A2})$$

i.e. the seasonal innovation is Q_γ on the first dummy and 0 on the remaining $s - 2$ dummies. Componentwise, the evolution is

$$\alpha_t = \alpha_{t-1} + \beta_{t-1} + \omega_{\alpha,t}, \quad \omega_{\alpha,t} \sim \mathcal{N}(0, Q_\alpha), \quad (\text{A3})$$

$$\beta_t = \beta_{t-1} + \omega_{\beta,t}, \quad \omega_{\beta,t} \sim \mathcal{N}(0, Q_\beta), \quad (\text{A4})$$

$$\boldsymbol{\gamma}_t = S\boldsymbol{\gamma}_{t-1} + \mathbf{e}_1 \omega_{\gamma,t}, \quad \omega_{\gamma,t} \sim \mathcal{N}(0, Q_\gamma). \quad (\text{A5})$$

Equivalently, writing $\boldsymbol{\gamma}_t = (\gamma_{1,t}, \dots, \gamma_{s-1,t})^\top$,

$$\gamma_{1,t} = - \sum_{j=1}^{s-1} \gamma_{j,t-1} + \omega_{\gamma,t}, \quad (\text{A6})$$

$$\gamma_{j,t} = \gamma_{j-1,t-1}, \quad j = 2, \dots, s-1, \quad (\text{A7})$$

so the innovation enters only through $\gamma_{1,t}$.

The latent location entering the observation layer is

$$\mu_t = \alpha_t + \gamma_{1,t} = \alpha_t + \mathbf{e}_1^\top \boldsymbol{\gamma}_t. \quad (\text{A8})$$

We now seek an ancillary representation (Yu and Meng, 2011; Papaspiliopoulos et al., 2007) that reduces posterior dependence between the latent trajectory and the innovation scales. Write $Q_k = q_k^2$ with $q_k > 0$ and factorise

$$Q^{1/2} = \text{diag}(q_\alpha, q_\beta) \oplus (q_\gamma \mathbf{e}_1 \mathbf{e}_1^\top), \quad q_k = \sqrt{Q_k}. \quad (\text{A9})$$

A naive rescaling of $\mathbf{x}_t$ by $Q^{-1/2}$ would make the process noise unit-variance, but it also induces q -dependent state transition and therefore does not produce an ancillary state prior. Following Frühwirth-Schnatter and Wagner (2010), we instead construct a non-centred state whose transition law is free of q and move all dependence on q into the observation mapping; by adding an extra "slope-integrator" term in the state (and thus increasing the dimension as we wrote in 2.1).

From (A4) write $\omega_{\beta,t} = q_{\beta}\tilde{\omega}_{\beta,t}$ with $\tilde{\omega}_{\beta,t} \sim \mathcal{N}(0, 1)$ and define

$$\tilde{\beta}_t := \sum_{j=1}^t \tilde{\omega}_{\beta,j}, \quad \tilde{\beta}_0 = 0, \quad \tilde{\beta}_t = \tilde{\beta}_{t-1} + \tilde{\omega}_{\beta,t}, \quad (\text{A10})$$

so that

$$\beta_t = \beta_0 + q_{\beta}\tilde{\beta}_t. \quad (\text{A11})$$

Similarly write $\omega_{\alpha,t} = q_{\alpha}\tilde{\omega}_{\alpha,t}$ with $\tilde{\omega}_{\alpha,t} \sim \mathcal{N}(0, 1)$ and define

$$\tilde{\alpha}_t := \sum_{j=1}^t \tilde{\omega}_{\alpha,j}, \quad \tilde{\alpha}_0 = 0, \quad \tilde{\alpha}_t = \tilde{\alpha}_{t-1} + \tilde{\omega}_{\alpha,t}, \quad (\text{A12})$$

$$A_t := \sum_{j=1}^t \tilde{\beta}_{j-1}, \quad A_0 = 0, \quad A_t = A_{t-1} + \tilde{\beta}_{t-1}, \quad (\text{A13})$$

where A_t indeed "integrates" the slopes. This yields the mapping

$$\alpha_t = \underbrace{\alpha_0 + t\beta_0}_{\text{deterministic trend}} + \underbrace{q_{\alpha}\tilde{\alpha}_t + q_{\beta}A_t}_{\text{stochastic deviations}}. \quad (\text{A14})$$

From (A5), write the seasonal innovation as

$$\mathbf{e}_1 \omega_{\gamma,t} = q_{\gamma} \mathbf{e}_1 \tilde{\omega}_{\gamma,t}, \quad \tilde{\omega}_{\gamma,t} \sim \mathcal{N}(0, 1), \quad (\text{A15})$$

and define the unit-variance seasonal deviation state by

$$\tilde{\gamma}_t = S\tilde{\gamma}_{t-1} + \mathbf{e}_1 \tilde{\omega}_{\gamma,t}, \quad \tilde{\gamma}_0 = \mathbf{0}. \quad (\text{A16})$$

Unrolling (A16) gives $\tilde{\gamma}_t = \sum_{j=1}^t S^{t-j} \mathbf{e}_1 \tilde{\omega}_{\gamma,j}$ and hence

$$\gamma_t = S^t \gamma_0 + q_{\gamma} \tilde{\gamma}_t. \quad (\text{A17})$$

Taking the active component,

$$\gamma_{1,t} = \mathbf{e}_1^{\top} S^t \gamma_0 + q_{\gamma} \mathbf{e}_1^{\top} \tilde{\gamma}_t = \underbrace{\gamma_{0,t}}_{\text{baseline season}} + q_{\gamma} \tilde{\gamma}_{1,t}, \quad (\text{A18})$$

where

$$\gamma_{0,t} := \mathbf{e}_1^{\top} S^t \gamma_0. \quad (\text{A19})$$

Combining (A8), (A14) and (A18) yields

$$\mu_t = \underbrace{\alpha_0 + t\beta_0 + \gamma_{0,t}}_{=:c_t} + q_{\alpha}\tilde{\alpha}_t + q_{\beta}A_t + q_{\gamma}\tilde{\gamma}_{1,t}. \quad (\text{A20})$$

Collect the non-centred states in the enlarged $(s+2)$ -dimensional vector

$$\tilde{\mathbf{x}}_t = (\tilde{\alpha}_t, \tilde{\beta}_t, A_t, \tilde{\gamma}_t^\top)^\top \in \mathbb{R}^{s+2}, \quad \tilde{\mathbf{x}}_0 = \mathbf{0}. \quad (\text{A21})$$

From (A12)–(A16), the evolution is

$$\tilde{\mathbf{x}}_t = \tilde{G} \tilde{\mathbf{x}}_{t-1} + \tilde{\omega}_t, \quad \tilde{\omega}_t \sim \mathcal{N}(\mathbf{0}, \tilde{Q}), \quad (\text{A22})$$

with

$$\tilde{G} = \begin{pmatrix} 1 & 0 & 0 & \mathbf{0}_{1 \times (s-1)} \\ 0 & 1 & 0 & \mathbf{0}_{1 \times (s-1)} \\ 0 & 1 & 1 & \mathbf{0}_{1 \times (s-1)} \\ \mathbf{0}_{(s-1) \times 1} & \mathbf{0}_{(s-1) \times 1} & \mathbf{0}_{(s-1) \times 1} & S \end{pmatrix}, \quad (\text{A23})$$

and innovation covariance

$$\tilde{Q} = \text{diag}(1, 1, 0, 1, 0, \dots, 0) = \text{diag}(1, 1, 0) \oplus (\mathbf{e}_1 \mathbf{e}_1^\top), \quad (\text{A24})$$

i.e. only $(\tilde{\alpha}_t, \tilde{\beta}_t, \tilde{\gamma}_{1,t})$ receive unit-variance innovations, while A_t and the remaining seasonal dummy coordinates (as before) evolve deterministically given the previous state.

All dependence on $(q_\alpha, q_\beta, q_\gamma)$ has been removed from the state evolution; it enters only through the mapping from $\tilde{\mathbf{x}}_t$ to the location μ_t . Define

$$c_t := \alpha_0 + t\beta_0 + \gamma_{0,t}, \quad (\text{A25})$$

and the design row

$$H(Q) = (q_\alpha, 0, q_\beta, q_\gamma \mathbf{e}_1^\top), \quad (\text{A26})$$

so that

$$\mu_t = c_t + H(Q) \tilde{\mathbf{x}}_t, \quad (\text{A27})$$

hence the label "non-centred". The observation equation is still

$$y_t \mid \tilde{\mathbf{x}}_t, q, \boldsymbol{\eta} \sim p(y_t \mid \mu_t, \boldsymbol{\eta}), \quad (\text{A28})$$

but now with another mapping from the non-centred states to the location parameter.

Finally, the centred structural components can always be recovered from the non-centred states:

$$\beta_t = \beta_0 + q_\beta \tilde{\beta}_t, \quad (\text{A29})$$

$$\alpha_t = \alpha_0 + t\beta_0 + q_\alpha \tilde{\alpha}_t + q_\beta A_t, \quad (\text{A30})$$

$$\gamma_t = S^t \gamma_0 + q_\gamma \tilde{\gamma}_t, \quad \gamma_{1,t} = \gamma_{0,t} + q_\gamma \tilde{\gamma}_{1,t}. \quad (\text{A31})$$

Substituting (A30) and (A31) into $\mu_t = \alpha_t + \gamma_{1,t}$ yields (A20), hence the centred and non-centred parametrisations induce the same distribution for $(\mu_1, \dots, \mu_T)$ and therefore for the observed data under (A28).

Appendix B Forward Filtering Backward Simulation

In the linear–Gaussian case, the full conditional distribution of the entire latent state trajectory is multivariate Gaussian and can be sampled exactly using the forward–filtering backward–sampling (FFBS) simulation smoother of [Carter and Kohn \(1994\)](#).

Forward filtering.

Let $(m_{t-1|t-1}, P_{t-1|t-1})$ denote the filtered mean and covariance of $\mathbf{x}_{t-1}$ given $y_{1:t-1}$. The one–step–ahead prediction moments are

$$m_{t|t-1} = Gm_{t-1|t-1}, \quad P_{t|t-1} = GP_{t-1|t-1}G^\top + Q. \quad (\text{B32})$$

Define the innovation, its variance, and the Kalman gain by

$$v_t = y_t - Hm_{t|t-1}, \quad S_t = HP_{t|t-1}H^\top + R_t, \quad K_t = P_{t|t-1}H^\top S_t^{-1}. \quad (\text{B33})$$

The filtered moments $(m_{t|t}, P_{t|t})$ then follow as

$$m_{t|t} = m_{t|t-1} + K_tv_t, \quad (\text{B34})$$

$$P_{t|t} = P_{t|t-1} - K_tS_tK_t^\top. \quad (\text{B35})$$

Backward simulation.

Define the smoothing gain

$$J_t = P_{t|t}G^\top P_{t+1|t}^{-1}, \quad t = 0, \dots, T-1. \quad (\text{B36})$$

FFBS produces a single exact draw from $p(\mathbf{x}_{0:T} \mid y_{1:T})$. First draw

$$\mathbf{x}_T \sim \mathcal{N}(m_T, P_T). \quad (\text{B37})$$

Then, for $t = T-1, \dots, 0$, sample from the Gaussian backward kernel

$$\mathbf{x}_t \mid \mathbf{x}_{t+1}, y_{1:T} \sim \mathcal{N}\left(m_{t|t} + J_t(\mathbf{x}_{t+1} - m_{t+1|t}), P_{t|t} - J_tP_{t+1|t}J_t^\top\right). \quad (\text{B38})$$

Appendix C Laplace approximation of the DGEV

In this appendix we derive a local Laplace approximation of the DGEV observation density which can be used as the basis for a Carter–Kohn simulation smoother [Carter and Kohn \(1994\)](#), by turning the non-Gaussian observation equation into a time–varying linear–Gaussian pseudo–observation ([Durbin and Koopman, 2000](#)).

We denote the log-density of a GEV at time t as

$$\ell_t(\mu_t) := \log g(y_t \mid \mu_t; \sigma, \xi).$$

For $\xi \neq 0$, define

$$z_t(\mu_t) = \frac{y_t - \mu_t}{\sigma}, \quad u_t(\mu_t) = 1 + \xi z_t(\mu_t),$$

so that the support condition is $u_t(\mu_t) > 0$. Then

$$\ell_t(\mu_t) = -\log \sigma - \left(1 + \frac{1}{\xi}\right) \log u_t(\mu_t) - (u_t(\mu_t))^{-1/\xi}. \quad (\text{C39})$$

We expand $\ell_t(\mu_t)$ to second order around a current value μ_t^* (e.g. a prediction $Gm_{t|t-1}$ from the previous Kalman step),

$$\ell_t(\mu_t) \approx \ell_t(\mu_t^*) + \ell_t'(\mu_t^*)(\mu_t - \mu_t^*) + \frac{1}{2} \ell_t''(\mu_t^*)(\mu_t - \mu_t^*)^2. \quad (\text{C40})$$

From (C39) one obtains

$$\frac{\partial z_t}{\partial \mu_t} = -\frac{1}{\sigma}, \quad \frac{\partial u_t}{\partial \mu_t} = \xi \frac{\partial z_t}{\partial \mu_t} = -\frac{\xi}{\sigma}.$$

By the chain rule,

$$\frac{\partial}{\partial \mu_t} \log u_t(\mu_t) = \frac{1}{u_t(\mu_t)} \frac{\partial u_t}{\partial \mu_t} = -\frac{\xi}{\sigma} \frac{1}{u_t(\mu_t)}, \quad (\text{C41})$$

$$\frac{\partial}{\partial \mu_t} (u_t(\mu_t))^{-1/\xi} = -\frac{1}{\xi} (u_t(\mu_t))^{-1/\xi-1} \frac{\partial u_t}{\partial \mu_t} = \frac{1}{\sigma} (u_t(\mu_t))^{-1/\xi-1}. \quad (\text{C42})$$

Substituting into (C39) gives

$$\ell_t'(\mu_t) = \frac{\partial}{\partial \mu_t} \ell_t(\mu_t) = \frac{1}{\sigma} \left(\frac{\xi + 1}{u_t(\mu_t)} - (u_t(\mu_t))^{-1/\xi-1} \right). \quad (\text{C43})$$

Differentiating (C43) once more yields

$$\begin{aligned} \frac{\partial^2}{\partial \mu_t^2} \ell_t(\mu_t) &= \frac{1}{\sigma} \left[(\xi + 1) \frac{\partial}{\partial \mu_t} (u_t^{-1}) - \frac{\partial}{\partial \mu_t} (u_t^{-1/\xi-1}) \right] \\ &= \frac{1}{\sigma} \left[(\xi + 1) \left(-u_t^{-2} \frac{\partial u_t}{\partial \mu_t} \right) - \left(-\left(\frac{1}{\xi} + 1\right) u_t^{-1/\xi-2} \frac{\partial u_t}{\partial \mu_t} \right) \right] \\ &= \frac{1}{\sigma} \left[(\xi + 1) \left(-u_t^{-2} \left(-\frac{\xi}{\sigma} \right) \right) - \left(-\left(\frac{1}{\xi} + 1\right) u_t^{-1/\xi-2} \left(-\frac{\xi}{\sigma} \right) \right) \right] \\ &= \frac{1}{\sigma^2} \left[\xi(\xi + 1) u_t^{-2} - (\xi + 1) u_t^{-1/\xi-2} \right]. \end{aligned} \quad (\text{C44})$$

Evaluated at the expansion point μ_t^* , we write

$$u_t^* := u_t(\mu_t^*), \quad \ell_t'^* := \ell_t'(\mu_t^*), \quad \ell_t''^* := \ell_t''(\mu_t^*).$$

Now a Gaussian observation model with observation z_t and variance R_t has log-density

$$\log \mathcal{N}(z_t \mid \mu_t, R_t) = -\frac{1}{2} \log(2\pi R_t) - \frac{1}{2R_t} (z_t - \mu_t)^2.$$

Expanding this in μ_t around the same μ_t^* gives

$$\log \mathcal{N}(z_t \mid \mu_t, R_t) = C + \frac{1}{R_t} (z_t - \mu_t^*)(\mu_t - \mu_t^*) - \frac{1}{2R_t} (\mu_t - \mu_t^*)^2, \quad (\text{C45})$$

for some constant C not depending on μ_t .

Comparing coefficients of $(\mu_t - \mu_t^*)$ and $(\mu_t - \mu_t^*)^2$ between (C40) and the Gaussian expansion, we obtain

$$\ell_t'^* = \frac{1}{R_t} (z_t - \mu_t^*), \quad \ell_t''^* = -\frac{1}{R_t}. \quad (\text{C46})$$

Thus

$$R_t = -\frac{1}{\ell_t''^*}, \quad z_t = \mu_t^* + R_t \ell_t'^*. \quad (\text{C47})$$

The Laplace approximation above thus induces a linear-Gaussian "pseudo-observation" model,

$$z_t = H\mathbf{x}_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, R_t), \quad (\text{C48})$$

with z_t and R_t as in (C47). This allows us to apply the Carter-Kohn simulation smoother, as if the observations were Gaussian. Concretely, one such iteration (i), at time t proceeds as follows :

1. **Prediction step.** Given $(m_{t-1|t-1}, P_{t-1|t-1})$, compute

$$m_{t|t-1} = Gm_{t-1|t-1}, \quad P_{t|t-1} = GP_{t-1|t-1}G^\top + Q,$$

and set $\mu_t^* = \mu_t^{(i-1)}$.

2. **Local Laplace approximation.** Using μ_t^* , evaluate z_t and R_t via (C47). This defines the pseudo-observation model (C48).
3. **Update step (Kalman).** Compute the Kalman gain

$$S_t = HP_{t|t-1}H^\top + R_t, \quad K_t = P_{t|t-1}H^\top S_t^{-1},$$

and the updated mean and covariance

$$m_{t|t} = m_{t|t-1} + K_t(z_t - Hm_{t|t-1}), \quad P_{t|t} = P_{t|t-1} - K_t S_t K_t^\top.$$

Appendix D Full conditional distributions

This appendix collects the full conditional distributions needed for our Gibbs scheme in Section 3 when working in the non-centred parametrisation (NCP). However, remember that we can always map back to the original parametrization for reporting purposes. We show how to update the NCP state trajectories $\tilde{\mathbf{x}}_{0:T}$ and static parameter vectors $\boldsymbol{\theta}$ according to the scheme (19) - (20).

D.1 Latent state trajectory

The state full conditional is, up to proportionality,

$$p(\tilde{\mathbf{x}}_{0:T} \mid y_{1:T}, \boldsymbol{\theta}) \propto \prod_{t=1}^T \mathcal{N}(\tilde{\mathbf{x}}_t \mid \tilde{G}\tilde{\mathbf{x}}_{t-1}, \tilde{Q}) \prod_{t=1}^T p(y_t \mid c_t + H(Q)\tilde{\mathbf{x}}_t, \boldsymbol{\eta}). \quad (\text{D49})$$

In the DLM, $p(y_t \mid \mu_t, \sigma^2) = \mathcal{N}(y_t \mid \mu_t, \sigma^2)$, hence (D49) is multivariate Gaussian and we draw $\tilde{\mathbf{x}}_{0:T}$ exactly using FFBS (Appendix B), with observation row $H(Q)$ and offset c_t . In the DGEV, $p(y_t \mid \mu_t, \sigma, \xi)$ is GEV - we draw a state trajectory by applying the local Laplace pseudo-observation construction of Appendix C.

D.2 State-layer parameters

The baseline parameters $\alpha_0, \beta_0, \gamma_0$ and innovation scales $q_\alpha, q_\beta, q_\gamma$ are treated on the same level in the NCP. Therefore we combine them into one $s + 4$ -dimensional regression block

$$\boldsymbol{\delta} = (\alpha_0, \beta_0, \gamma_0^\top, q_\alpha, q_\beta, q_\gamma)^\top \quad (\text{D50})$$

with Gaussian prior (see (21) and (22)), so that

$$\boldsymbol{\theta} = (\boldsymbol{\delta}, \boldsymbol{\eta}) \quad (\text{D51})$$

more compactly.

Define the seasonal baseline row

$$\mathbf{b}_t^\top := \mathbf{e}_1^\top S^t \in \mathbb{R}^{1 \times (s-1)}, \quad (\text{D52})$$

and extract from $\tilde{\mathbf{x}}_t$ the three "deviation regressors"

$$u_{\alpha,t} := \tilde{\alpha}_t, \quad u_{\beta,t} := A_t, \quad u_{\gamma,t} := \tilde{\gamma}_{1,t}. \quad (\text{D53})$$

Then

$$\mu_t = \underbrace{\begin{pmatrix} 1 & t & \mathbf{b}_t^\top & u_{\alpha,t} & u_{\beta,t} & u_{\gamma,t} \end{pmatrix}}_{=:X_t} \underbrace{\begin{pmatrix} \alpha_0 \\ \beta_0 \\ \gamma_0 \\ q_\alpha \\ q_\beta \\ q_\gamma \end{pmatrix}}_{=\boldsymbol{\delta}}. \quad (\text{D54})$$

Stacking rows X_t gives a design matrix $X \in \mathbb{R}^{T \times (2+(s-1)+3)}$.

In the DLM we have $\mathbf{y} \mid \boldsymbol{\delta}, \tilde{\mathbf{x}}_{1:T}, \sigma^2 \sim \mathcal{N}(X\boldsymbol{\delta}, \sigma^2 I_T)$. With a Gaussian prior $\boldsymbol{\delta} \sim \mathcal{N}(\mathbf{m}_\delta, V_\delta)$, the full conditional is again Gaussian,

$$\boldsymbol{\delta} \mid y_{1:T}, \tilde{\mathbf{x}}_{1:T}, \sigma^2 \sim \mathcal{N}(\mathbf{m}_\delta^*, V_\delta^*), \quad (\text{D55})$$

$$V_\delta^{*-1} = V_\delta^{-1} + \frac{1}{\sigma^2} X^\top X, \quad (\text{D56})$$

$$\mathbf{m}_\delta^* = V_\delta^* \left(V_\delta^{-1} \mathbf{m}_\delta + \frac{1}{\sigma^2} X^\top \mathbf{y} \right), \quad (\text{D57})$$

with $\mathbf{y} = (y_1, \dots, y_T)^\top$.

Similarly for the DGEV under the Laplace approximation of Appendix C, we have $\mathbf{z} \mid \boldsymbol{\delta}, \tilde{\mathbf{x}}_{1:T}, (\sigma, \xi) \approx \mathcal{N}(X\boldsymbol{\delta}, R)$ with $R = \text{diag}(R_1, \dots, R_T)$ and response $\mathbf{z} = (z_1, \dots, z_T)^\top$. Conditional on the pseudo data $(\mathbf{z}, R)$, we now have the approximate update step

$$\boldsymbol{\delta} \mid z_{1:T}, R_{1:T}, \tilde{\mathbf{x}}_{1:T} \sim \mathcal{N}(\mathbf{m}_\delta^*, V_\delta^*), \quad (\text{D58})$$

$$V_\delta^{*-1} = V_\delta^{-1} + X^\top R^{-1} X, \quad (\text{D59})$$

$$\mathbf{m}_\delta^* = V_\delta^* (V_\delta^{-1} \mathbf{m}_\delta + X^\top R^{-1} \mathbf{z}). \quad (\text{D60})$$

D.2.1 Bayesian lasso hyperparameter updates for q

We impose the hierarchical Bayesian lasso prior (21) (Belmonte et al., 2014) on each innovation standard deviation q_k . This is a shrinkage prior in the sense that it prefers small values but keeps heavy tails so that genuinely time-varying components can still pick up large q_k . The q_k are updated in the joint regression block (D50) (together with the deterministic baseline parameters); conditional on the current q_k , the local scales τ_k and the global scale λ in (21) admit conjugate updates as we'll show below.

Local scales

The full conditional of τ_k is

$$p(\tau_k \mid q_k, \lambda^2) \propto p(q_k \mid \tau_k) p(\tau_k \mid \lambda^2).$$

Using $p(q_k | \tau_k) \propto \tau_k^{-1/2} \exp\{-q_k^2/(2\tau_k)\}$ and $p(\tau_k | \lambda^2) \propto \exp\{-(\lambda^2/2)\tau_k\}$ yields

$$p(\tau_k | q_k, \lambda^2) \propto \tau_k^{-1/2} \exp\left\{-\frac{1}{2} \left(\frac{q_k^2}{\tau_k} + \lambda^2 \tau_k\right)\right\}, \quad \tau_k > 0, \quad (\text{D61})$$

which is a generalized inverse Gaussian distribution,

$$\tau_k | q_k, \lambda^2 \sim \text{GIG}(p = \frac{1}{2}, a = \lambda^2, b = q_k^2). \quad (\text{D62})$$

A convenient sampling form follows by the reparameterisation $u_k := 1/\tau_k$. By change of variables in (D61),

$$p(u_k | q_k, \lambda^2) \propto u_k^{-3/2} \exp\left\{-\frac{1}{2} \left(q_k^2 u_k + \frac{\lambda^2}{u_k}\right)\right\}, \quad u_k > 0,$$

which matches the inverse Gaussian family. In particular,

$$u_k | q_k, \lambda^2 \sim \text{InvGaussian}\left(\frac{\lambda}{|q_k|}, \lambda^2\right) \quad (\text{D63})$$

This is the standard Bayesian-lasso augmentation used for Gibbs sampling (Park and Casella, 2008), which avoids numerical overflow in the inverse-Gaussian mean.

Global scale

Combining

$$p(\tau_k | \lambda^2) = \frac{\lambda^2}{2} \exp\left(-\frac{\lambda^2}{2} \tau_k\right),$$

over $k = 1, \dots, 3$ with the Gamma prior $\lambda^2 \sim \text{Gamma}(a_\lambda, b_\lambda)$ gives

$$p(\lambda^2 | \tau_{1:3}) \propto (\lambda^2)^{a_\lambda-1} \exp(-b_\lambda \lambda^2) \prod_{k=1}^3 \left(\lambda^2 \exp\left\{-\frac{\lambda^2}{2} \tau_k\right\} \right),$$

hence the conjugate Gamma full conditional

$$\lambda^2 | \tau_{1:3} \sim \text{Gamma}\left(a_\lambda + 3, b_\lambda + \frac{1}{2} \sum_{k=1}^3 \tau_k\right). \quad (\text{D64})$$

We then set $\lambda = \sqrt{\lambda^2}$.

D.3 Random sign switches

Because μ_t depends on $(q_k, \tilde{x}_{0:T,k})$ ($\tilde{x}_{t,k}$ being the q_k corresponding NCP-coordinate at time t) only through the products $q_k \tilde{x}_{k,t}$, the posterior is invariant under simultaneous random sign flips

$$q_k \leftarrow -q_k, \quad \tilde{x}_{0:T,k} \leftarrow -\tilde{x}_{0:T,k} \quad (\text{D65})$$

Applying these moves after each iteration improves mixing in the innovation-scale parameters for slowly varying components (Frühwirth-Schnatter and Wagner, 2010; Belmonte et al., 2014; Frühwirth-Schnatter, 2001)

D.4 Observation-layer parameters

The observation parameter conditional factorises over t :

$$p(\boldsymbol{\eta} \mid y_{1:T}, \tilde{\mathbf{x}}_{0:T}, \boldsymbol{\delta}) \propto p(\boldsymbol{\eta}) \prod_{t=1}^T p(y_t \mid \mu_t, \boldsymbol{\eta}). \quad (\text{D66})$$

DLM: σ^2

With prior $\sigma^2 \sim \text{IG}(a_\sigma, b_\sigma)$ this is conjugate,

$$\sigma^2 \mid y_{1:T}, \tilde{\mathbf{x}}_{0:T}, \boldsymbol{\delta} \sim \text{IG}\left(a_\sigma + \frac{T}{2}, b_\sigma + \frac{1}{2} \sum_{t=1}^T (y_t - \mu_t)^2\right). \quad (\text{D67})$$

DGEV: (σ, ξ)

With priors $\sigma^2 \sim \text{IG}(a_\sigma, b_\sigma)$ and $\xi \sim U(\xi_{\min}, \xi_{\max})$, we have

$$p(\sigma, \xi \mid y_{1:T}, \tilde{\mathbf{x}}_{0:T}, \boldsymbol{\delta}) \propto p(\sigma^2) p(\xi) \prod_{t=1}^T g(y_t \mid \mu_t; \sigma, \xi), \quad (\text{D68})$$

which we update using a random-walk Metropolis–Hastings step (and reject proposals that violate the GEV support constraints).

Appendix E Evolution of exceedance probabilities under a Weibull tail

Let Y_t denote a block maximum at time t with $Y_t \sim \text{GEV}(\mu_t, \sigma, \xi)$, where $\sigma > 0$ and ξ are fixed while μ_t varies over time. For $\xi \neq 0$ the CDF is

$$G_t(y) = \exp\left\{-u_t(y)^{-1/\xi}\right\}, \quad u_t(y) := 1 + \xi \frac{y - \mu_t}{\sigma} > 0, \quad (\text{E69})$$

and the exceedance probability of a threshold y^* equals

$$p_t(y^*) := \Pr(Y_t > y^* \mid \mu_t, \sigma, \xi) = 1 - G_t(y^*). \quad (\text{E70})$$

When $\xi < 0$ (Weibull domain of attraction), the support is bounded above by the finite right endpoint

$$y_{\max, t} = \mu_t - \frac{\sigma}{\xi} = \mu_t + \frac{\sigma}{|\xi|}, \quad (\text{E71})$$

so $p_t(y^*) \equiv 0$ whenever $y^* \geq y_{\max, t}$.

Define the endpoint margin

$$\varepsilon_t(y^*) := y_{\max, t} - y^*. \quad (\text{E72})$$

For $\xi < 0$ we can rewrite the support term as

$$u_t(y^*) = \frac{|\xi|}{\sigma} \varepsilon_t(y^*), \quad (\text{E73})$$

and substituting into (E69)–(E70) yields

$$p_t(y^*) = \mathbf{1}\{\varepsilon_t(y^*) > 0\} \left[1 - \exp\left\{-\left(\frac{|\xi|}{\sigma} \varepsilon_t(y^*)\right)^{1/|\xi|}\right\} \right]. \quad (\text{E74})$$

i.e., the threshold becomes possible precisely when $\varepsilon_t(y^*)$ turns positive (i.e. once the moving endpoint exceeds y^*).

Over short windows, approximate the location by a (local) linear trend with fixed slope and intercept

$$\mu_t \approx \mu_{t_0} + \beta(t - t_0), \quad (\text{E75})$$

so the endpoint and margin evolve in parallel,

$$y_{\max, t} \approx y_{\max, t_0} + \beta(t - t_0), \quad \varepsilon_t(y^*) \approx \varepsilon_{t_0}(y^*) + \beta(t - t_0). \quad (\text{E76})$$

If we now take t_0 to be the crossing time: $y_{\max, t_0} = y^*$, equivalently $\varepsilon_{t_0}(y^*) = 0$. For $\beta > 0$ and $t > t_0$, substituting $\varepsilon_t(y^*) \approx \beta(t - t_0)$ into (E74) gives

$$p_t(y^*) \approx 1 - \exp\left\{-\left(\frac{|\xi|}{\sigma} \beta(t - t_0)\right)^{1/|\xi|}\right\}, \quad t > t_0. \quad (\text{E77})$$

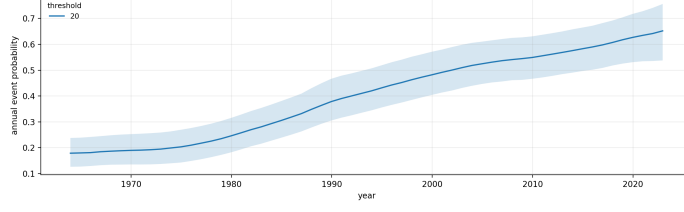


Fig. F1: (Annual) tropical-night risk from the TNx DGEV model. With exceedance probabilities on the y-axis instead of return periods.

As $t \downarrow t_0$ the exponent is small, hence to leading order,

$$p_t(y^*) \sim \left(\frac{|\xi|}{\sigma} \beta (t - t_0) \right)^{1/|\xi|} \quad (t \downarrow t_0, \beta > 0). \quad (\text{E78})$$

Thus, once the endpoint has crossed the threshold, exceedance probability increases (briefly) as a power law with exponent $1/|\xi|$.

Appendix F Empirical results

Table F1: Posterior medians (90% credible intervals) for selected latent states not shown in the main text. Levels α_t are in $^{\circ}\text{C}$ and slopes β_t in $^{\circ}\text{C}/\text{decade}$.

Series	Year	α_t ($^{\circ}\text{C}$)	β_t ($^{\circ}\text{C}/\text{decade}$)
TXn	1892	6.62 [5.91, 7.23]	0.28 [0.11, 0.46]
TXn	1950	7.90 [7.53, 8.30]	0.20 [-0.19, 0.55]
TXn	2020	9.99 [9.42, 10.59]	0.34 [-0.12, 0.90]
TNn	1892	0.23 [-0.54, 0.85]	0.26 [0.08, 0.46]
TNn	1950	1.08 [0.67, 1.46]	0.21 [-0.14, 0.53]
TNn	2020	3.06 [2.46, 3.60]	0.26 [-0.23, 0.74]
TNx	1892	10.85 [10.49, 11.23]	0.05 [-0.40, 0.41]
TNx	1950	11.13 [10.90, 11.37]	0.04 [-0.32, 0.27]
TNx	2020	12.90 [12.50, 13.27]	0.27 [-0.14, 0.75]

This appendix collects some results that didn't make it to the main-text. Tables F1–F2 report posterior medians and 90% credible intervals for selected latent states (levels and slopes at representative years) and for innovation standard deviations (on the $\log_{10}$ scale), respectively. Figure F1 provides an additional risk-tracking example for warm nights by showing the posterior trajectory of the annual exceedance probability of the tropical-night threshold (see 8.1). We further show posterior histograms in F.1, forecasts in F.3 and cross-validation results in F.4.

Table F2: Posterior medians (90% credible intervals) for innovation standard deviations on the $\log_{10}$ scale.

Series	$\log_{10} q_\alpha $	$\log_{10} q_\beta $	$\log_{10} q_\gamma $
TXx	-1.38 [-1.92, -1.02]	-3.68 [-4.47, -3.10]	-1.54 [-2.16, -1.14]
TXn	-1.37 [-1.93, -0.94]	-4.07 [-4.96, -3.39]	-2.04 [-2.86, -1.45]
TNx	-1.77 [-2.46, -1.33]	-4.04 [-4.81, -3.49]	-1.51 [-1.83, -1.23]
TNn	-1.21 [-1.71, -0.81]	-3.89 [-4.71, -3.04]	-1.97 [-2.73, -1.38]
TXm	-1.378 [-2.017, -1.031]	-4.072 [-5.206, -3.462]	-2.092 [-3.206, -1.612]
TNm	-1.650 [-2.533, -1.273]	-4.170 [-5.046, -3.635]	-2.272 [-3.308, -1.775]

F.1 Posterior histograms

This section summarises marginal posterior uncertainty in key static parameters of the structural models. For the Gaussian DLM (TXm, TNm) we show histograms of the innovation variances $Q_\cdot$, together with the observation standard deviation σ in Figure F2. For the structural DGEV (TXx, TXn, TNx, TNn) we additionally show the GEV shape parameter ξ in Figure F3.

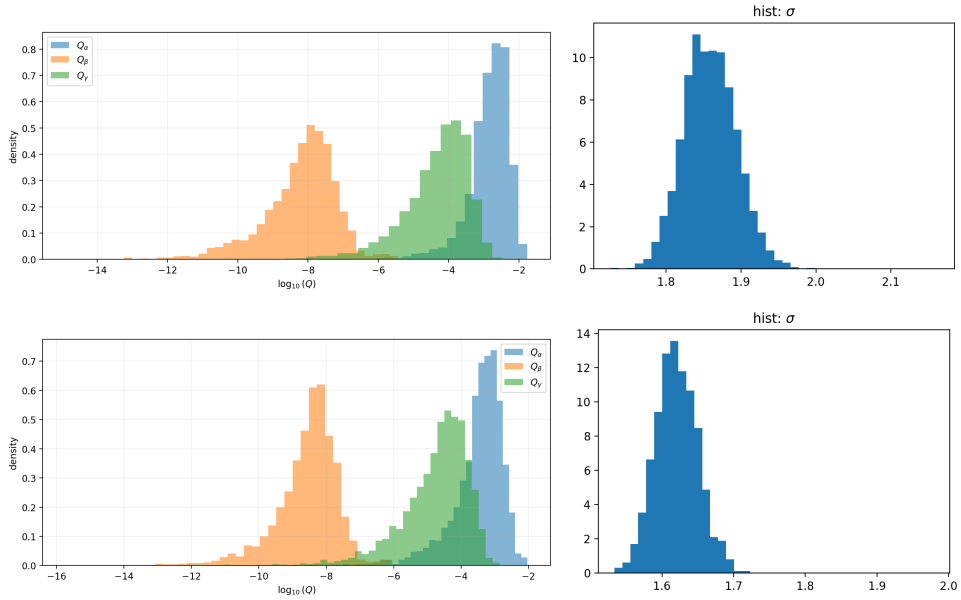


Fig. F2: Posterior histograms for the structural DLM fitted to monthly mean temperatures. Rows correspond to TXm (top) and TNm (bottom). The left column shows the posterior for the (logarithm of the) innovation variances $Q_\cdot$, and the right column shows the (observation) standard deviation σ .

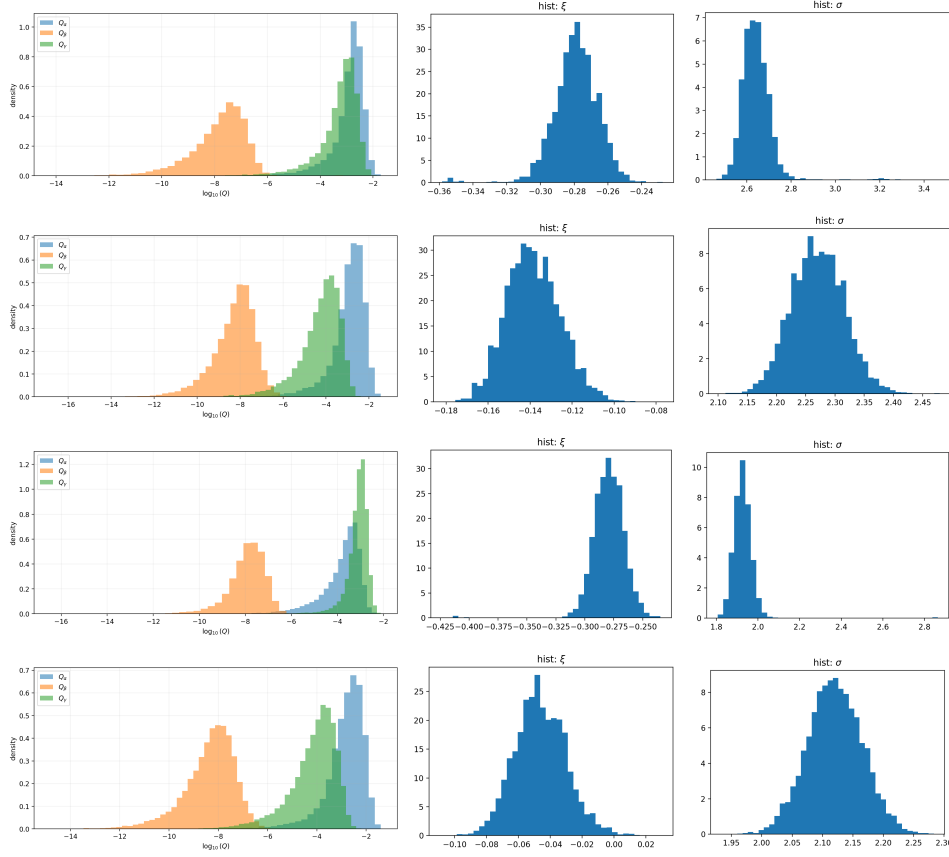


Fig. F3: Posterior histograms for the structural DGEV fitted to monthly temperature extremes. Rows correspond (top to bottom) to TXx, TXn, TNx and TNn. Columns show (left to right) the innovation variances Q_i , the GEV shape parameter ξ , and scale σ .

F.2 Seasonality

In Fig. F4 we show the seasonal pattern for all six temperature indices at three reference times (beginning, middle, end of the record; here 1892, 1957, 2022).

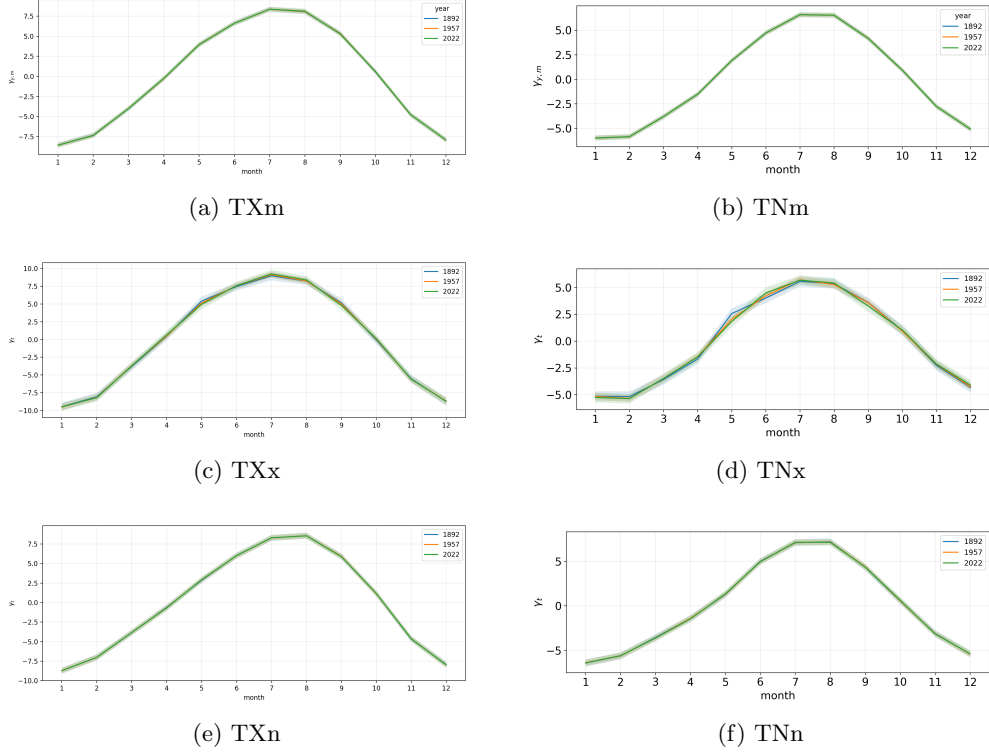


Fig. F4: Estimated seasonal patterns for monthly temperature summaries in Uccle. Left column: TX series; right column: TN series. Rows correspond to means (m), monthly maxima (x), and monthly minima (n). For each series the seasonal pattern is shown at three reference times (beginning, middle, end of the record) using posterior medians and 90% credible bands.

Two complementary views specifically for TNx are shown in Fig. F5 to illustrate how to interpret dynamically changing seasonal patterns. The left panel plots the posterior median seasonal contribution for each month as a function of calendar year (one curve per month), without uncertainty, which makes it slightly easier to spot potential slow drifts in specific months. The right panel displays the same information as a heatmap.

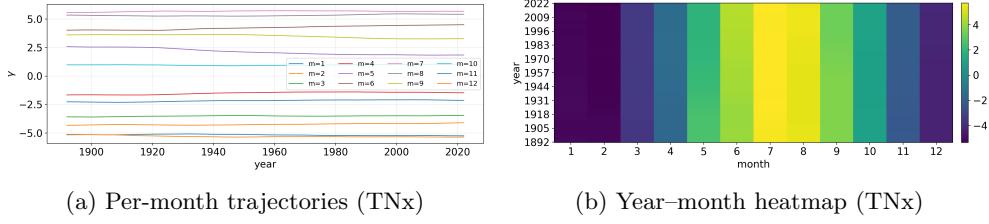


Fig. F5: TNx seasonal geometry. Left: posterior median seasonal contribution for each calendar month tracked through time. Right: the same information shown as a heatmap.

F.3 Forecasts

Figures F6–F7 show posterior predictions obtained by forward simulation from the fitted state-space models. The left column in each figure reports monthly forecasts, while the right column shows (aggregated) annual ones.

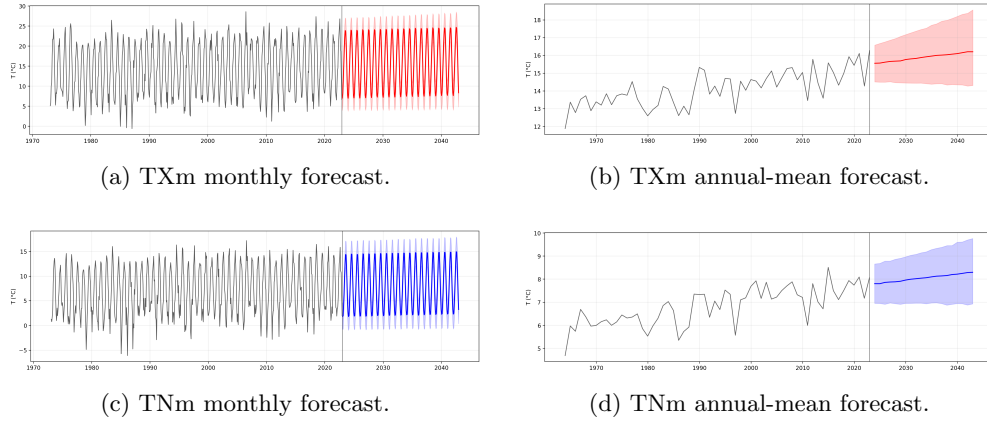


Fig. F6: Forecasts for monthly mean temperatures (left column) and annual means (right column). Top row: TXm; bottom row: TNm. Solid black denotes observations; coloured curve and shaded band denote median and 90% predictive interval.

F.4 Cross-validation

Predictive performance is assessed via an expanding-window cross-validation scheme. Figures F8–F10 display $H = 120$ -month (10-year) ahead posterior predictive forecast fans at split points 60%, 80%, and 90% of the record with held-out observations overlaid for visual comparison.

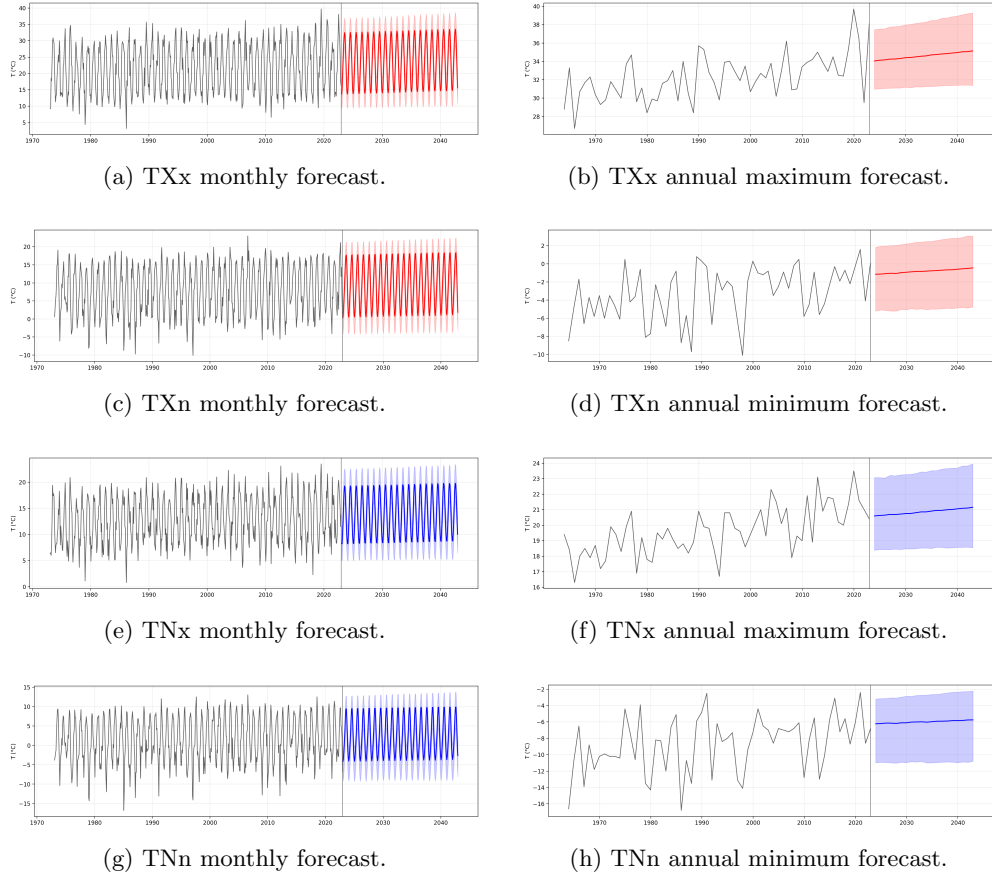


Fig. F7: Posterior predictive forecasts for monthly temperature extremes (left column) and annual extremes (right column). Rows correspond (top to bottom) to TXx, TXn, TNx, TNn. Solid black denotes observed monthly extremes; coloured curve and shaded band denote the median and 90% predictive interval.

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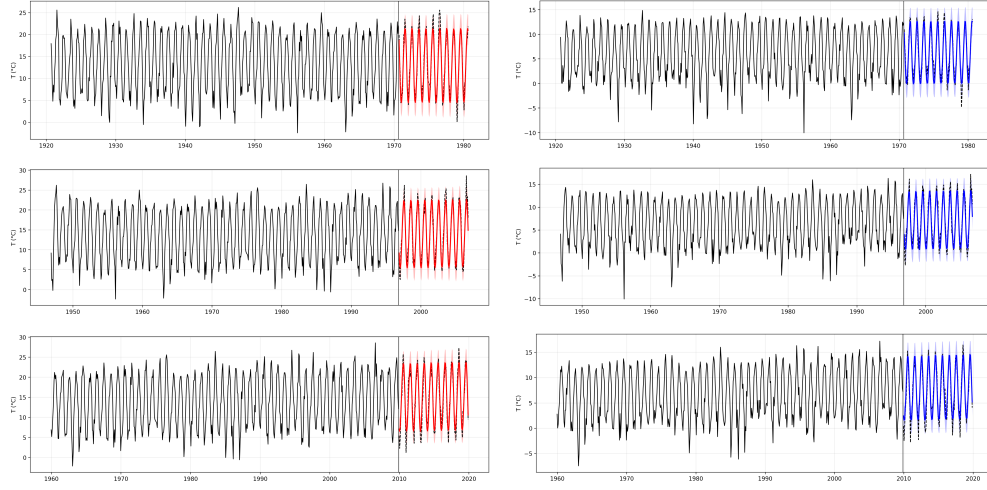


Fig. F8: Cross-validation forecasts for monthly mean temperatures (left: TXm; right: TNm) at split points 60%, 80%, and 90% of the record (top to bottom). Solid black: training data; dashed black: held-out observations; coloured curve and shaded band: posterior predictive median and 90% predictive interval. The vertical line marks the split time t_s .

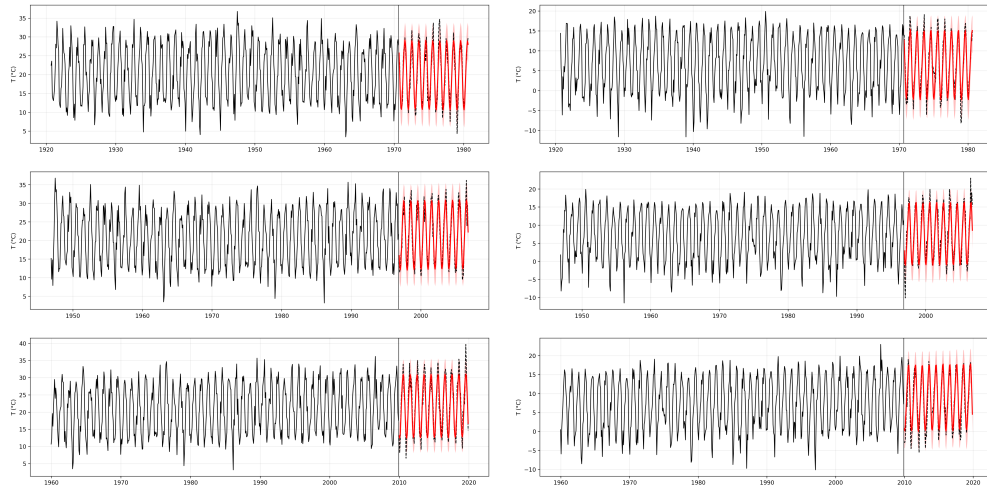


Fig. F9: Cross-validation forecasts for monthly TX extremes at split points 60%, 80%, and 90% of the record (top to bottom). Left column: TXx . Right column: TXn.

Alexander, L.V., Zhang, X., Peterson, T.C., Caesar, J., Gleason, B., Klein Tank, A., Haylock, M., Collins, D., Trewin, B., Rahimzadeh, F., et al.: Global observed changes in daily climate extremes of temperature and precipitation. *Journal of Geophysical Research: Atmospheres* **111**(D5) (2006)

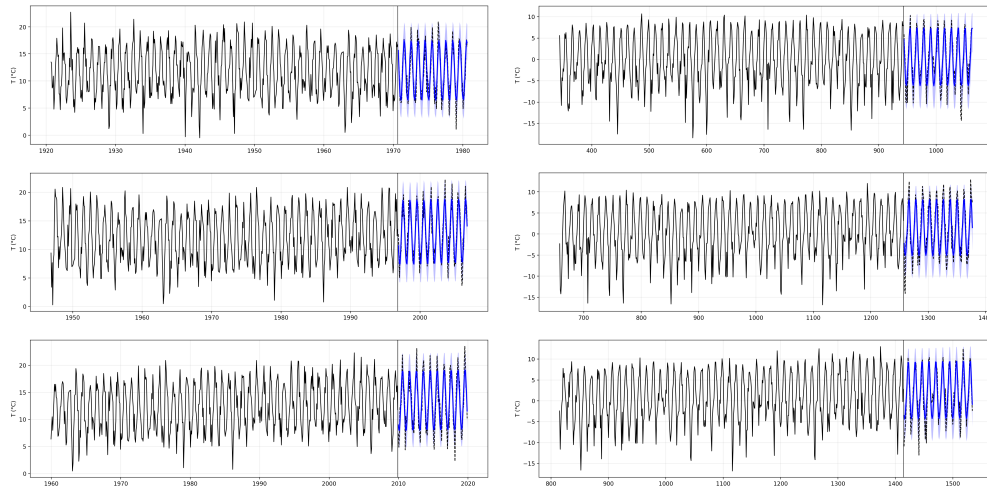


Fig. F10: Cross-validation forecasts for monthly TN extremes at split points 60%, 80%, and 90% of the record (top to bottom). Left column: TNx. Right column: TNn.

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